

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/switched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4

```

```
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)
```

```
return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8084
Nombre de lignes df clean	8084

Nombre de features	29
% de NaN	0.0
Taille dataset entraînement	5658
Taille dataset test	2425
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_12h_mean, r_NH3_st_12h_std, r_NH3_st_12h_rms, r_NH3_st_12h_median, r_NH3_st_12h_min, r_NH3_st_12h_max, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_temperature_st_12h_mean, r_temperature_st_12h_std, r_temperature_st_12h_rms, r_temperature_st_12h_median, r_temperature_st_12h_min, r_temperature_st_12h_max, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_umidity_st_12h_mean, r_umidity_st_12h_std, r_umidity_st_12h_rms, r_umidity_st_12h_median, r_umidity_st_12h_min, r_umidity_st_12h_max, r_umidity_st_12h_diff_last, r_umidity_st_12h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

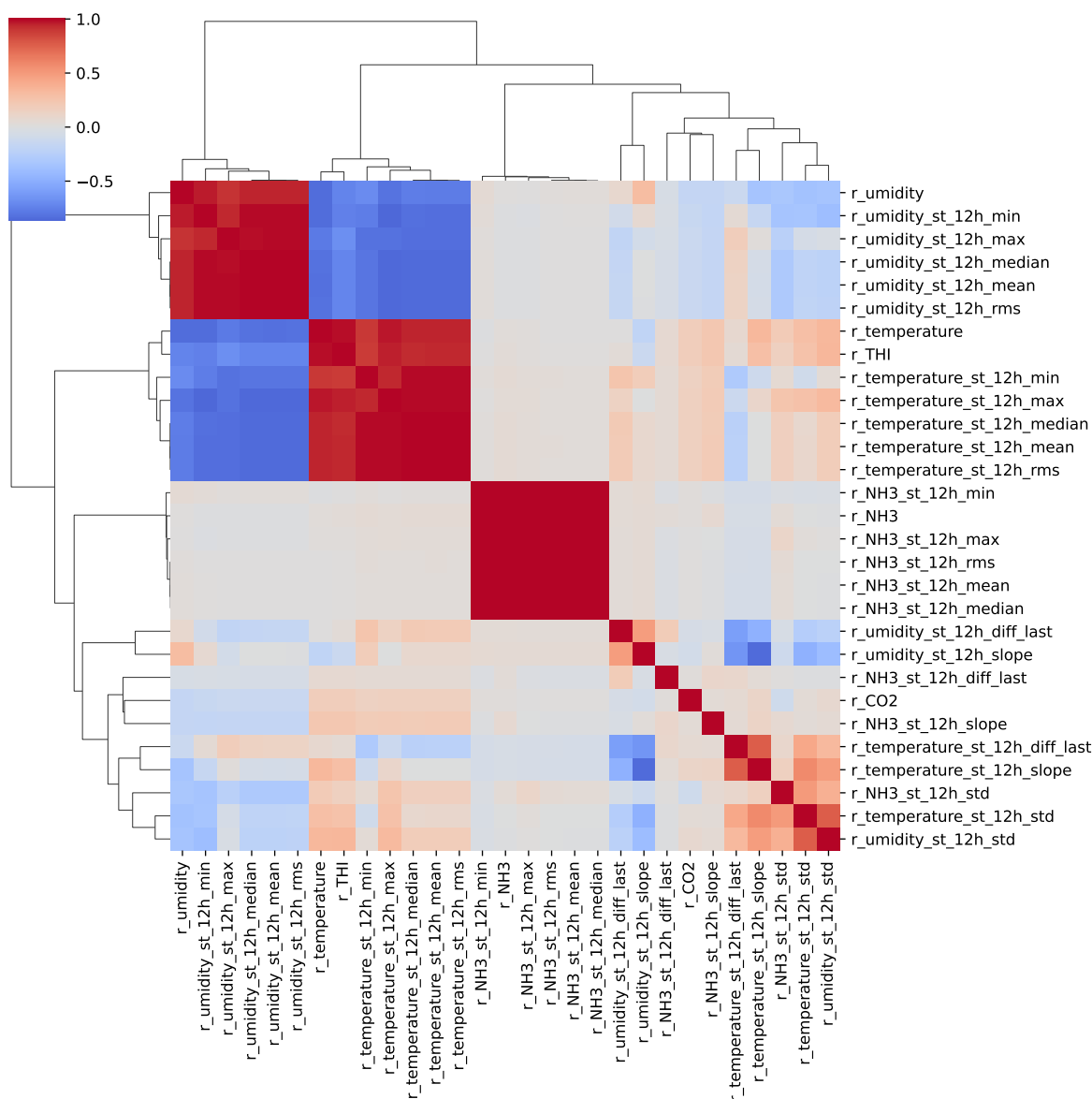
cluster 1: r_NH3_st_12h_rms, r_NH3_st_12h_mean, r_NH3_st_12h_median, r_NH3, r_NH3_st_12h_min, r_NH3_st_12h_max ; **cluster 2:** r_umidity_st_12h_max, r_temperature, r_umidity_st_12h_median, r_temperature_st_12h_rms, r_umidity_st_12h_min, r_temperature_st_12h_mean, r_temperature_st_12h_max, r_umidity_st_12h_rms, r_temperature_st_12h_median, r_THI, r_umidity_st_12h_mean, r_temperature_st_12h_min, r_umidity ; **cluster 3:** r_CO2 ; **cluster 4:** r_NH3_st_12h_std ; **cluster 5:** r_NH3_st_12h_diff_last ; **cluster 6:** r_NH3_st_12h_slope ; **cluster 7:** r_temperature_st_12h_std ; **cluster 8:** r_temperature_st_12h_diff_last

; **cluster 9:** r_temperature_st_12h_slope, r_umidity_st_12h_slope ; **cluster 10:** r_umidity_st_12h_std ; **cluster 11:** r_umidity_st_12h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3_st_12h_rms, r_umidity_st_12h_max, r_CO2, r_NH3_st_12h_std, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_temperature_st_12h_std, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_umidity_st_12h_std, r_umidity_st_12h_diff_last



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3_st_12h_rms, r_umidity_st_12h_max, r_CO2, r_NH3_st_12h_std, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_temperature_st_12h_std, r_temper-

ature_st_12h_diff_last, r_temperature_st_12h_slope, r_umidity_st_12h_std, r_umidity_st_12h_diff_last

Len(Features_X): 11

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

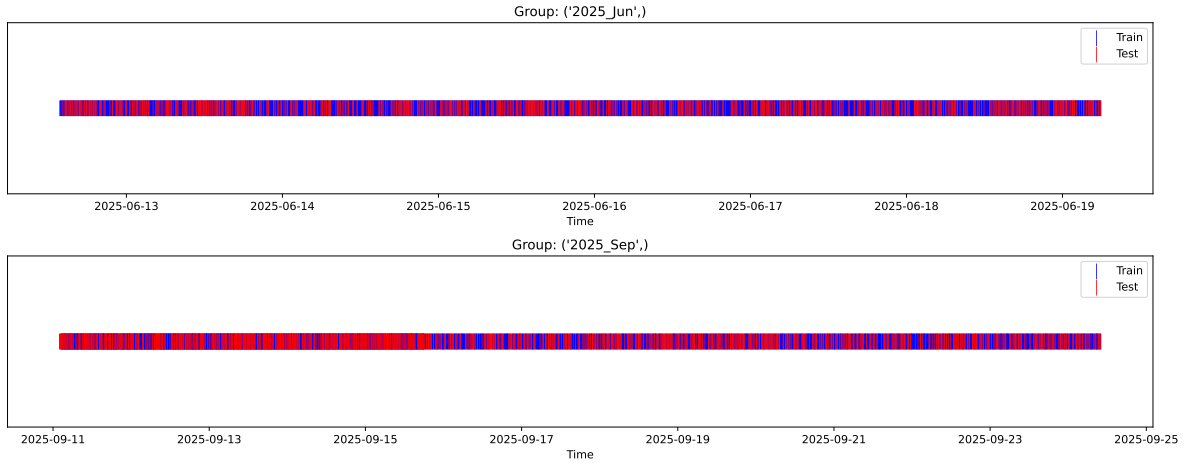
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

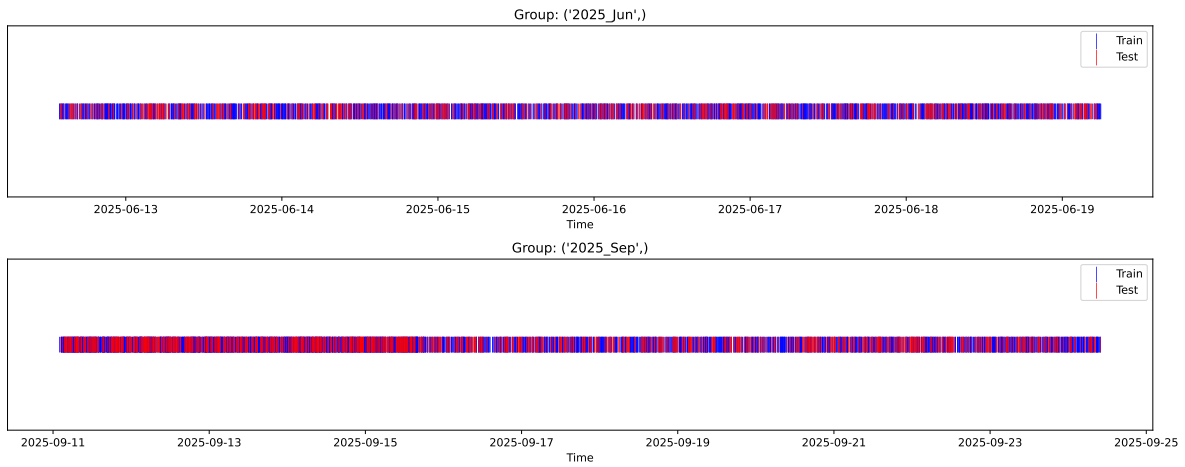
External Split: Train: 5658 samples Test: 2426 samples



3.3.2 Internal CV Folds

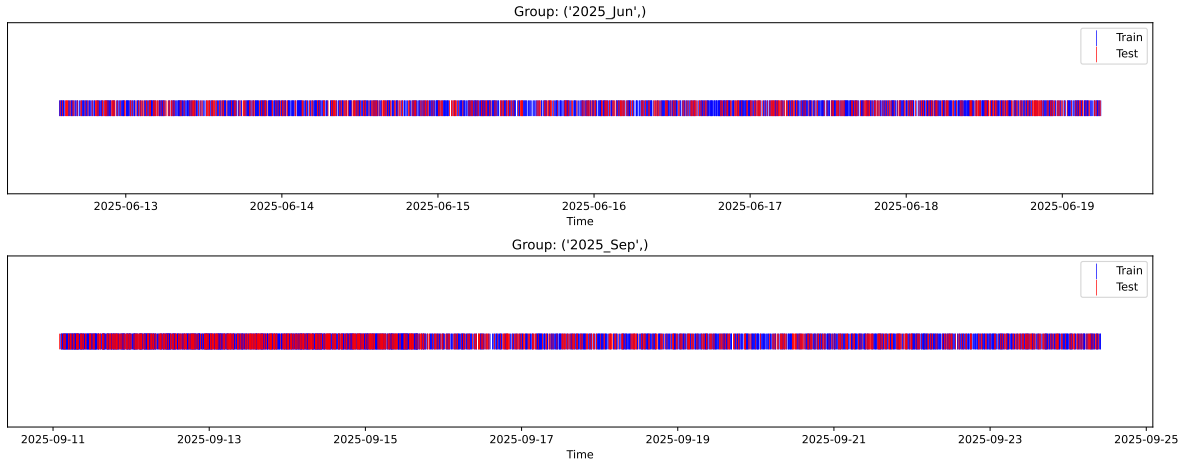
Fold 1/5

Train: 3960 samples Test: 1698 samples Excluded: 0 samples Window exclusion: 0h



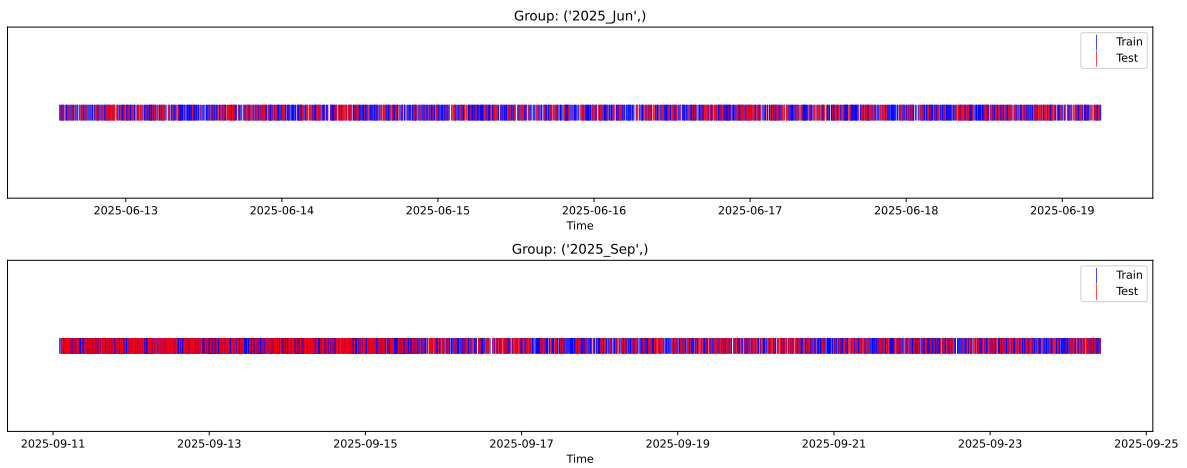
Fold 2/5

Train: 3960 samples Test: 1698 samples Excluded: 0 samples Window exclusion: 0h



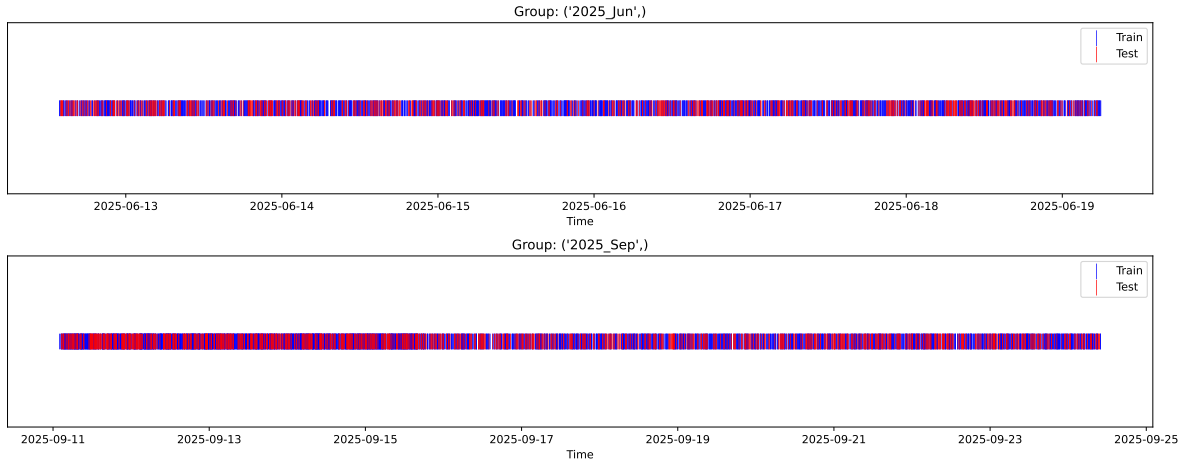
Fold 3/5

Train: 3960 samples Test: 1698 samples Excluded: 0 samples Window exclusion: 0h



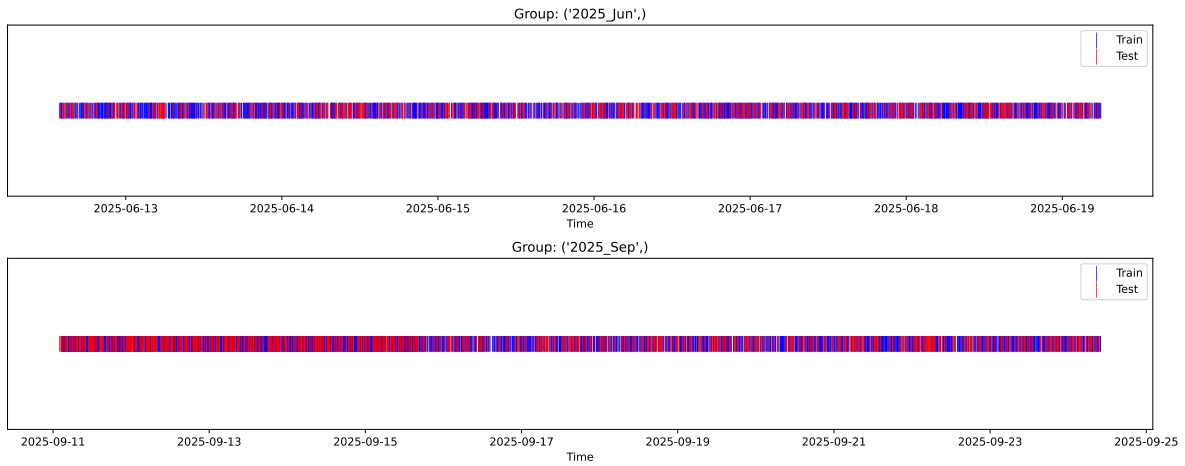
Fold 4/5

Train: 3960 samples Test: 1698 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3960 samples Test: 1698 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.346, std=0.796, min=0.959, max=13.845
 y_{test} : mean=2.362, std=0.797, min=0.973, max=7.779

Fold 0 — y_{test} : mean=2.35, std=0.84, y_{train} : mean=2.35, std=0.78

Fold 1 — y_{test} : mean=2.34, std=0.80, y_{train} : mean=2.35, std=0.80

Fold 2 — y_{test} : mean=2.33, std=0.82, y_{train} : mean=2.35, std=0.79

Fold 3 — y_test: mean=2.33, std=0.77, y_train: mean=2.35, std=0.81

Fold 4 — y_test: mean=2.32, std=0.75, y_train: mean=2.36, std=0.82

train scores CV: [0.54399173 0.5227705 0.53482496 0.52419569 0.5173882]

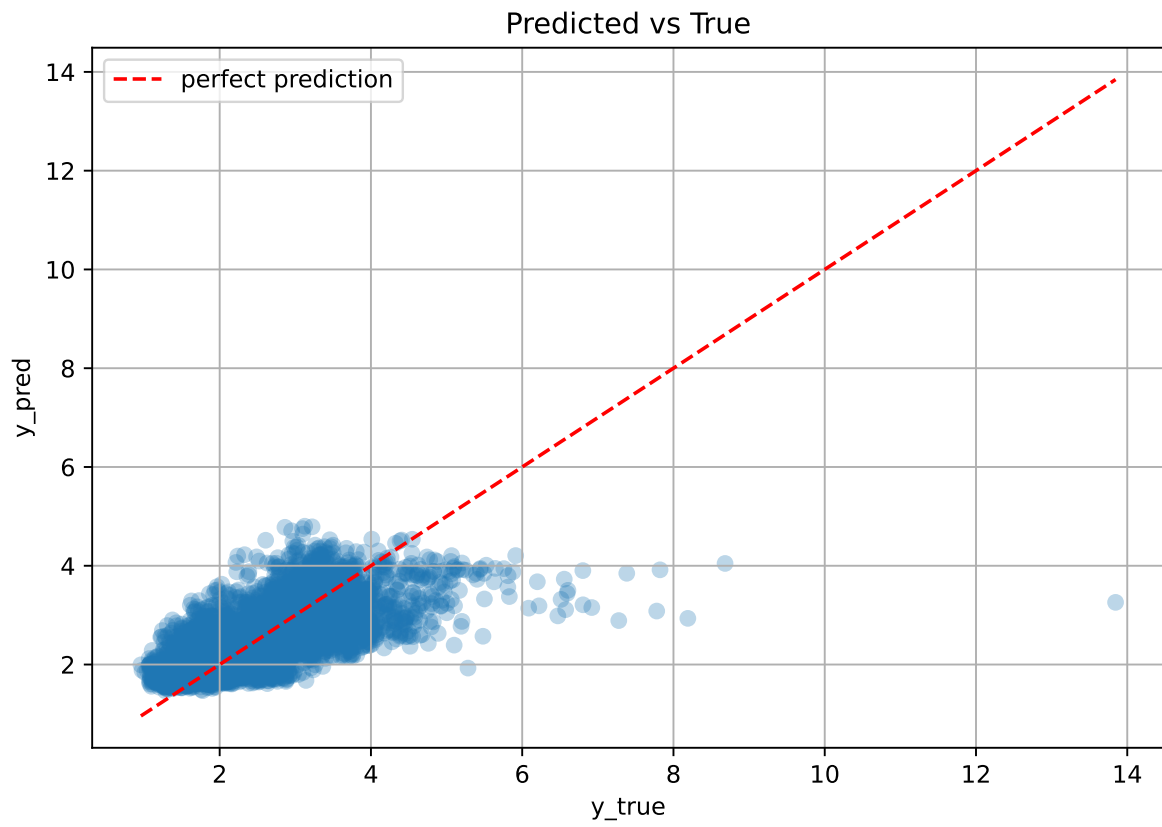
test scores CV: [0.48905648 0.53258892 0.5061125 0.52915901 0.54670596]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5286	0.5207	0.0205	0.5026	0.5029	-0.0178	0.9658	1.0152

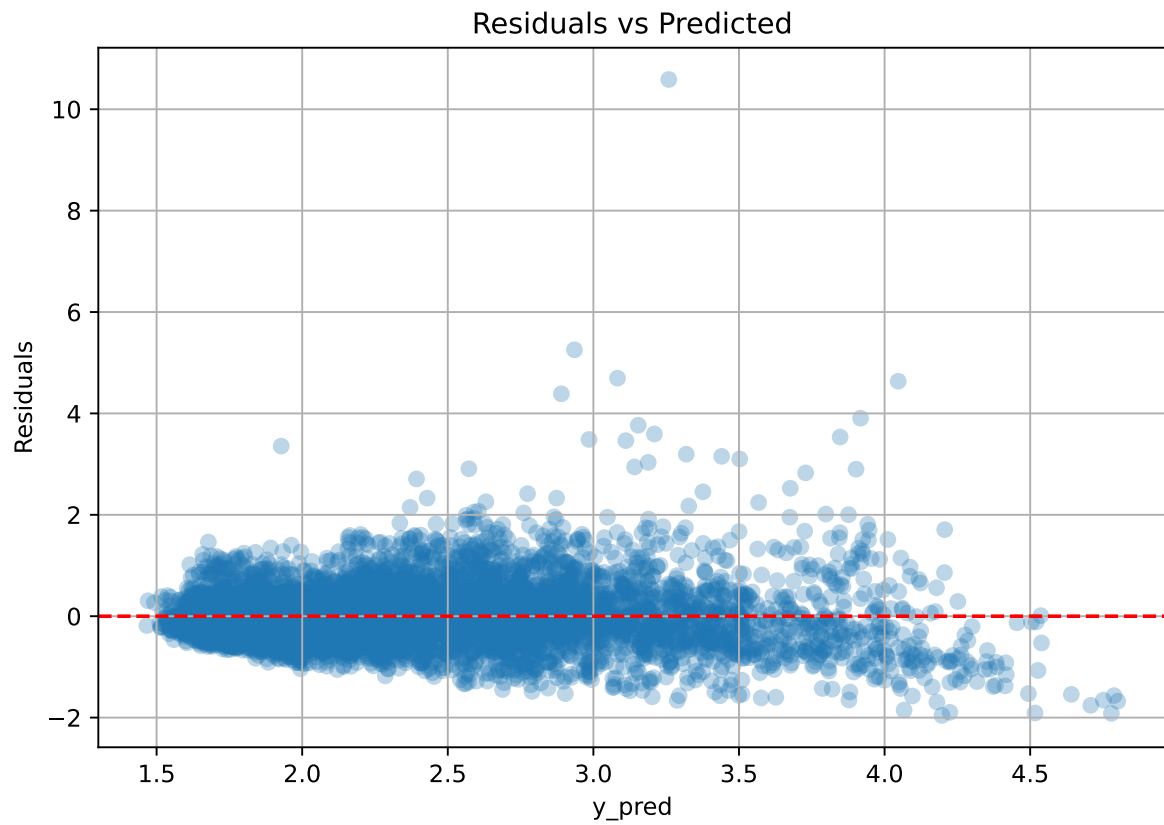
Coefficients:

coef_r_CO2 = -0.0374, coef_r_NH3_st_12h_diff_last = 0.0333, coef_r_NH3_st_12h_rms = -0.0387, coef_r_NH3_st_12h_slope = 0.0220, coef_r_NH3_st_12h_std = 0.0718, coef_r_temperature_st_12h_diff_last = 0.0569, coef_r_temperature_st_12h_slope = 0.2286, coef_r_temperature_st_12h_std = 0.1123, coef_r_umidity_st_12h_diff_last = -0.0946, coef_r_umidity_st_12h_max = -0.2770, coef_r_umidity_st_12h_std = 0.1061, intercept = 2.3463,

3.4.2 Scatter



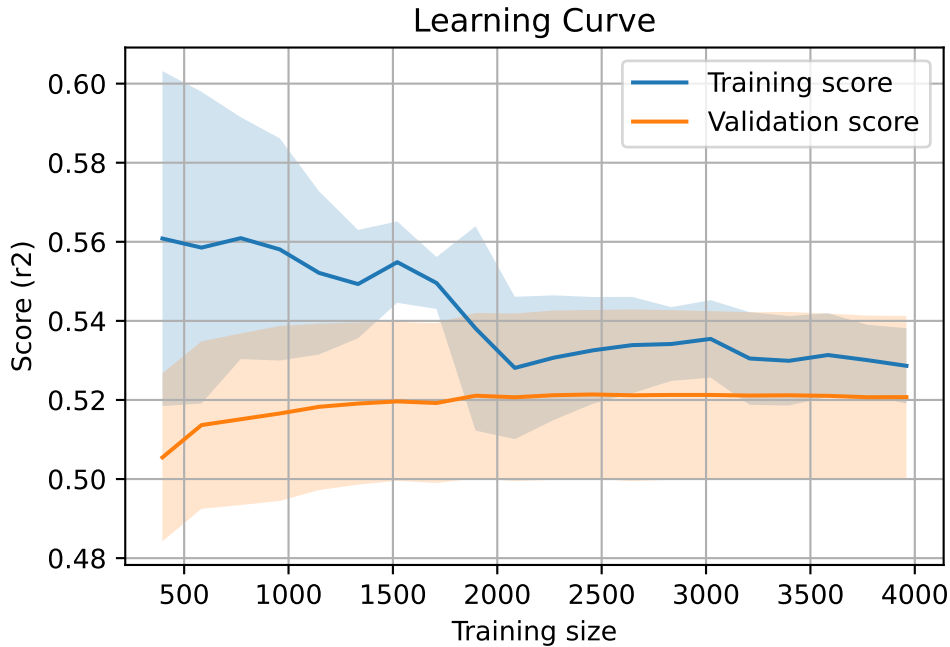
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 8084, **Len(training)** : 5659,

Len(training_real) : 3962, **Len(test_of_training)** : 1697, **Len(test)** : 2425

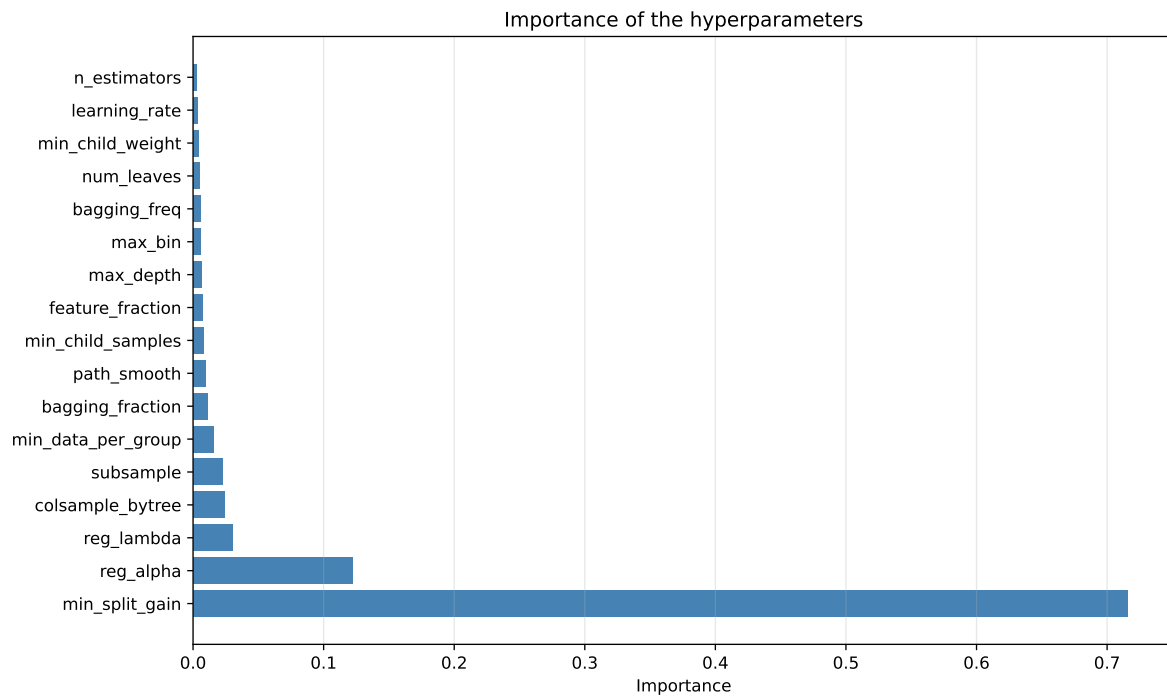
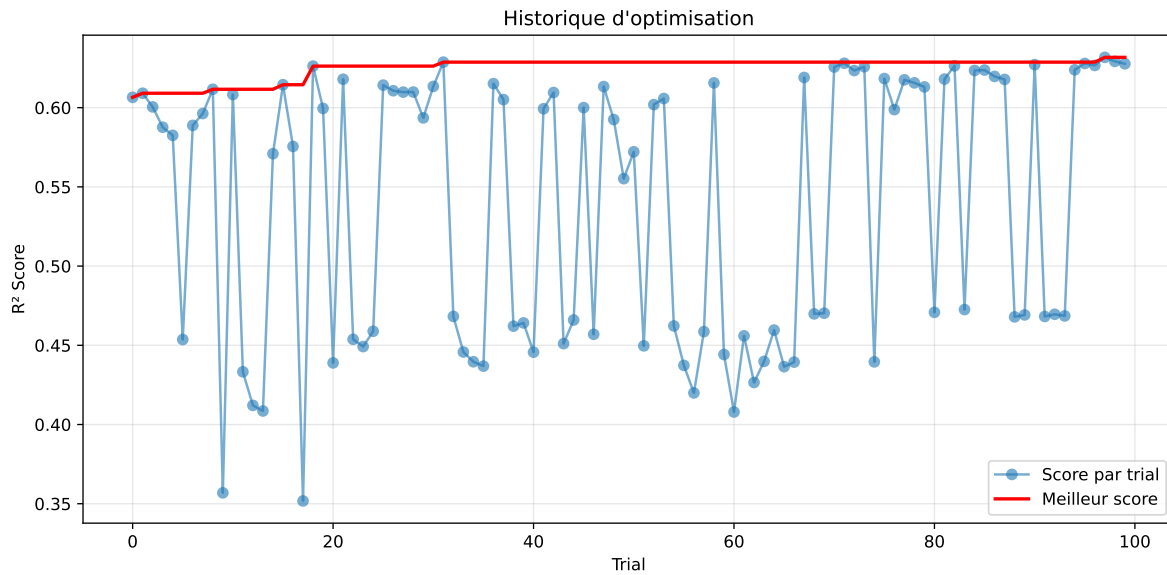


3.5 Model : LGBM

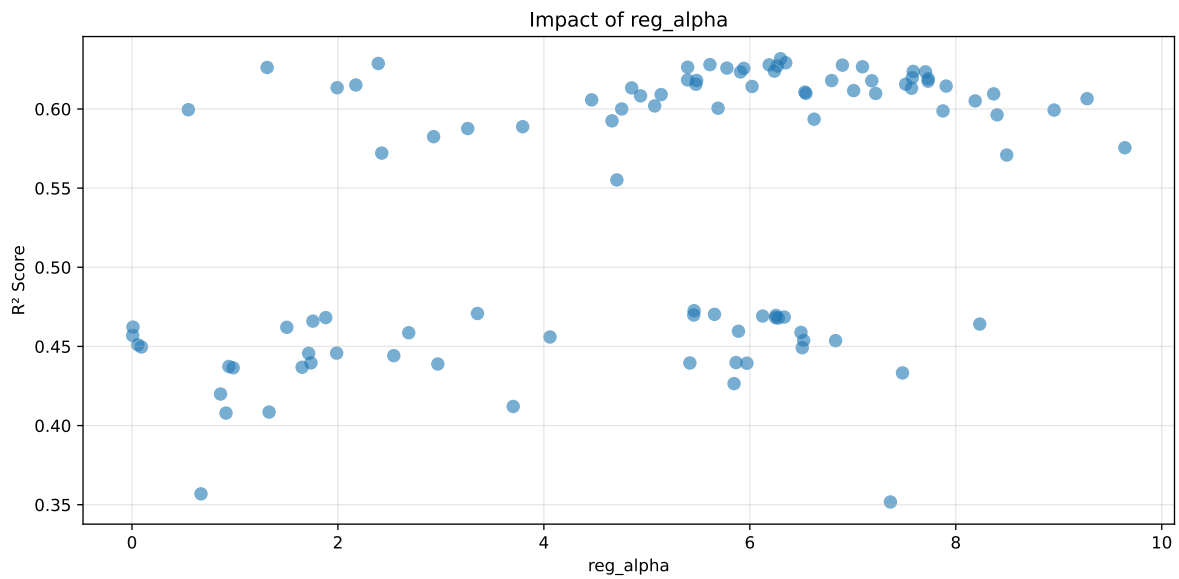
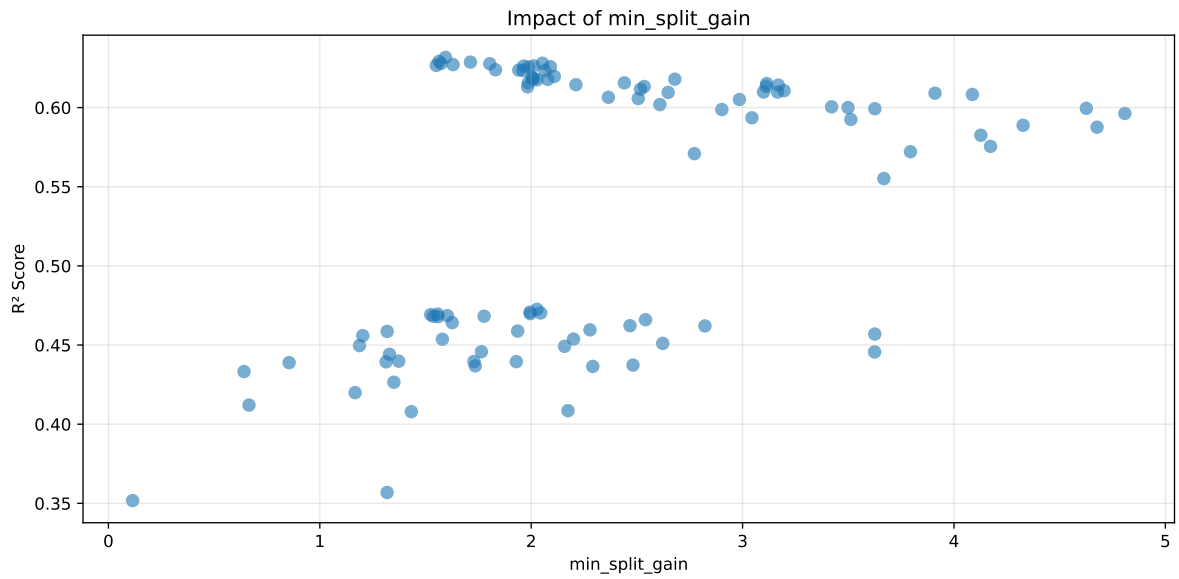
3.5.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.346, std=0.796, min=0.959, max=13.845
 y_{test} : mean=2.362, std=0.797, min=0.973, max=7.779

===== TOP Importance FEATURES =====
 feature importance r_CO2 427 r_umidity_st_12h_max 299 r_temperature_st_12h_slope
 183 r_NH3_st_12h_std 151 r_NH3_st_12h_rms 144 r_umidity_st_12h_std 143 r_temper-
 ature_st_12h_diff_last 112 r_NH3_st_12h_diff_last 88 r_umidity_st_12h_diff_last 87
 r_temperature_st_12h_std 68 r_NH3_st_12h_slope 58

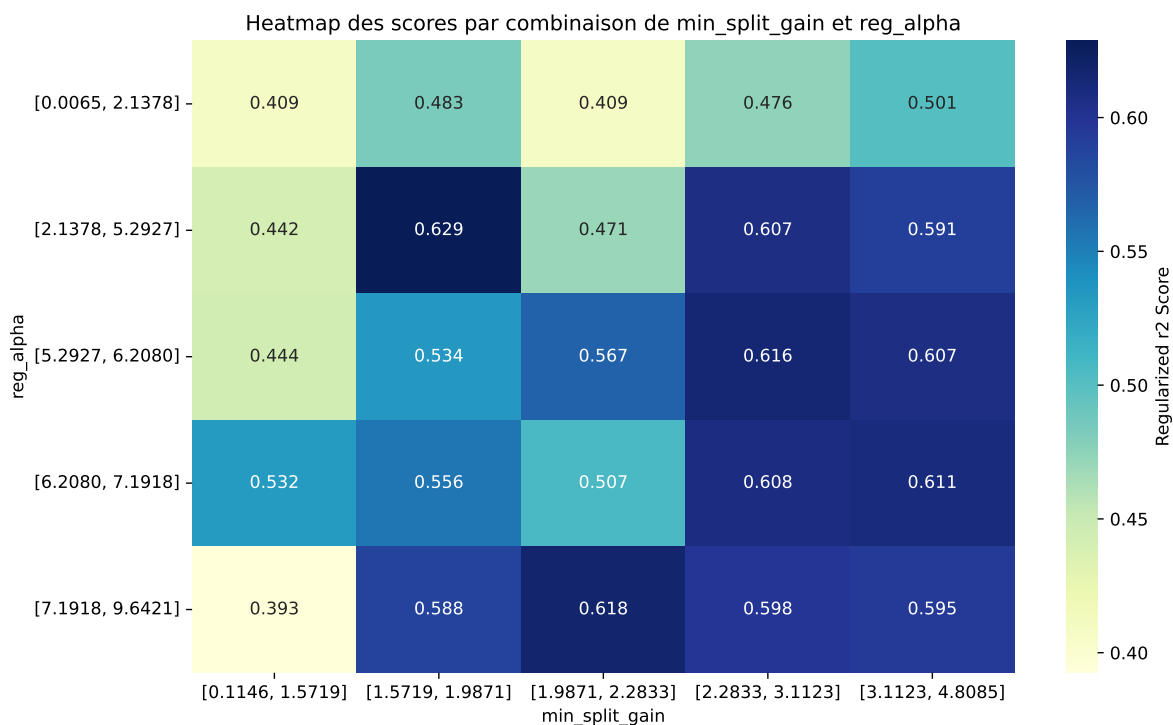


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of `observed=False` is deprecated and will change to `observed=True` in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6625	0.6318	0.6198	0.0245	0.6318	0.6377	0.006	1.0094	1.0486
2	0.6583	0.6292	0.616	0.0238	0.6292	0.6346	0.0054	1.0086	1.0462
3	0.66	0.6287	0.6162	0.0233	0.6287	0.6329	0.0041	1.0066	1.0497
4	0.6579	0.628	0.6147	0.0249	0.628	0.632	0.004	1.0064	1.0476
5	0.6576	0.6279	0.6154	0.0247	0.6279	0.6342	0.0063	1.01	1.0474
6	0.6557	0.6277	0.6128	0.0245	0.6277	0.6302	0.0025	1.004	1.0446
7	0.6579	0.6271	0.6158	0.0239	0.6271	0.6344	0.0073	1.0116	1.049
8	0.6562	0.6267	0.6141	0.0236	0.6267	0.6316	0.0049	1.0079	1.0471
9	0.6542	0.6264	0.6113	0.0247	0.6264	0.6299	0.0035	1.0056	1.0444
10	0.6565	0.6262	0.6142	0.0261	0.6262	0.6294	0.0031	1.005	1.0484
---	---	---	---	---	---	---	---	---	---
100	0.7492	0.6829	0.6762	0.0278	0.3517	0.6899	0.0069	1.0102	1.097

Hyperparameters:

Rank 1: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.023156540144267847, 'reg__num_leaves': 53, 'reg__subsample': 0.7720485394614578, 'reg__colsample_bytree': 0.7512504391432291, 'reg__min_child_samples': 36, 'reg__reg_alpha': 6.297929058520941, 'reg__reg_lambda': 11.10454073959811, 'reg__min_split_gain': 1.5946550946568783, 'reg__path_smooth': 0.7782618147344028, 'reg__min_child_weight':

5.24132266450585, 'reg__feature_fraction': 0.8459991510550603, 'reg__bagging_fraction': 0.5181917313399235, 'reg__bagging_freq': 1, 'reg__max_bin': 63, 'reg__min_data_per_group': 62}

Rank 2: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.023788092785304957, 'reg__num_leaves': 53, 'reg__subsample': 0.8052508475172696, 'reg__colsample_bytree': 0.7773577460372776, 'reg__min_child_samples': 36, 'reg__reg_alpha': 6.348708506245957, 'reg__reg_lambda': 17.487774257048514, 'reg__min_split_gain': 1.5638105496203853, 'reg__path_smooth': 0.8086559688750424, 'reg__min_child_weight': 2.4308682584453707, 'reg__feature_fraction': 0.7684971095482366, 'reg__bagging_fraction': 0.5177701630464495, 'reg__bagging_freq': 1, 'reg__max_bin': 67, 'reg__min_data_per_group': 60}

Rank 3: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.03231744601993397, 'reg__num_leaves': 44, 'reg__subsample': 0.5410389867687417, 'reg__colsample_bytree': 0.6261553163701203, 'reg__min_child_samples': 34, 'reg__reg_alpha': 2.3921333571838894, 'reg__reg_lambda': 31.623556099493392, 'reg__min_split_gain': 1.7125807471516463, 'reg__path_smooth': 0.22574350369457497, 'reg__min_child_weight': 0.2070133162400828, 'reg__feature_fraction': 0.7723073431199454, 'reg__bagging_fraction': 0.5040611384429198, 'reg__bagging_freq': 3, 'reg__max_bin': 130, 'reg__min_data_per_group': 156}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.02660506951801894, 'reg__num_leaves': 47, 'reg__subsample': 0.7159322308258291, 'reg__colsample_bytree': 0.7107317856768939, 'reg__min_child_samples': 40, 'reg__reg_alpha': 5.612619265271679, 'reg__reg_lambda': 11.268815235836888, 'reg__min_split_gain': 2.0531869123131976, 'reg__path_smooth': 4.817021255550248, 'reg__min_child_weight': 3.64115371588773, 'reg__feature_fraction': 0.8193978805490771, 'reg__bagging_fraction': 0.5176596745135065, 'reg__bagging_freq': 1, 'reg__max_bin': 101, 'reg__min_data_per_group': 122}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.023958790565780052, 'reg__num_leaves': 53, 'reg__subsample': 0.7614044312119747, 'reg__colsample_bytree': 0.7434524926916326, 'reg__min_child_samples': 38, 'reg__reg_alpha': 6.187631841238049, 'reg__reg_lambda': 18.091898876520244, 'reg__min_split_gain': 1.5739181327752187, 'reg__path_smooth': 4.351146019743004, 'reg__min_child_weight': 4.816816159814651, 'reg__feature_fraction': 0.6728384048049905, 'reg__bagging_fraction': 0.5147205726826372, 'reg__bagging_freq': 1, 'reg__max_bin': 112, 'reg__min_data_per_group': 182}

Rank 6: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.02785605676155448, 'reg__num_leaves': 53, 'reg__subsample': 0.7775603808492154, 'reg__colsample_bytree': 0.7697761216271258, 'reg__min_child_samples': 36, 'reg__reg_alpha': 6.898963210024613, 'reg__reg_lambda': 16.70601485766524, 'reg__min_split_gain': 1.8040350203376772, 'reg__path_smooth': 0.7850969533496448, 'reg__min_child_weight':

2.4625505239884937, 'reg__feature_fraction': 0.7664274447219221, 'reg__bagging_fraction': 0.5494280439346216, 'reg__bagging_freq': 1, 'reg__max_bin': 74, 'reg__min_data_per_group': 61}

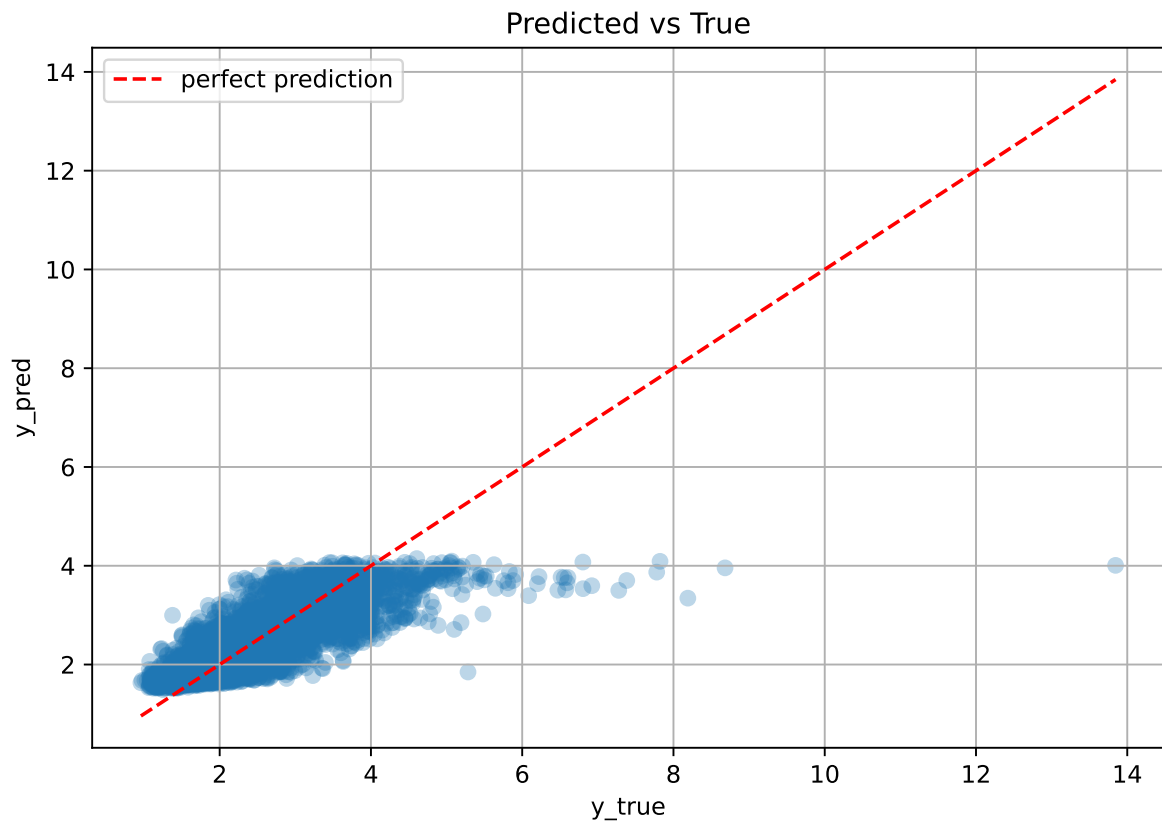
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.022646757688633844, 'reg__num_leaves': 49, 'reg__subsample': 0.7569693616127856, 'reg__colsample_bytree': 0.7440981513460293, 'reg__min_child_samples': 38, 'reg__reg_alpha': 6.266938924290529, 'reg__reg_lambda': 8.541351908953466, 'reg__min_split_gain': 1.6304804009162677, 'reg__path_smooth': 1.4849402577926574, 'reg__min_child_weight': 5.547114002779715, 'reg__feature_fraction': 0.669393522860057, 'reg__bagging_fraction': 0.5170493681235697, 'reg__bagging_freq': 1, 'reg__max_bin': 69, 'reg__min_data_per_group': 62}

Rank 8: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.023898503678425123, 'reg__num_leaves': 53, 'reg__subsample': 0.8100403680538127, 'reg__colsample_bytree': 0.753358714018951, 'reg__min_child_samples': 36, 'reg__reg_alpha': 7.094143864793576, 'reg__reg_lambda': 10.519337758730284, 'reg__min_split_gain': 1.5508663573210404, 'reg__path_smooth': 0.37456507317503523, 'reg__min_child_weight': 5.8934208110055595, 'reg__feature_fraction': 0.6688217010766964, 'reg__bagging_fraction': 0.5129045896724981, 'reg__bagging_freq': 1, 'reg__max_bin': 65, 'reg__min_data_per_group': 63}

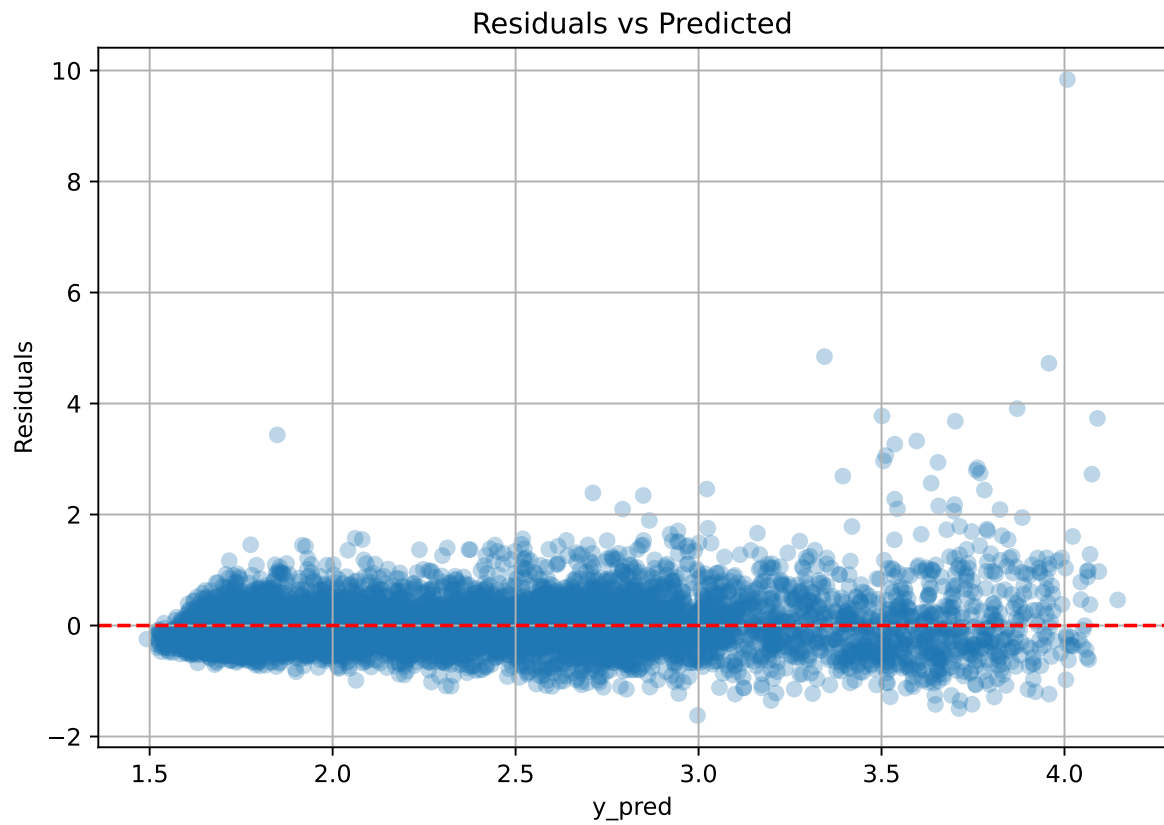
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.019043591372758684, 'reg__num_leaves': 59, 'reg__subsample': 0.7263193409842046, 'reg__colsample_bytree': 0.7085228634597529, 'reg__min_child_samples': 39, 'reg__reg_alpha': 5.396329349879877, 'reg__reg_lambda': 32.79884775471392, 'reg__min_split_gain': 2.0118090705990603, 'reg__path_smooth': 4.9698001125189, 'reg__min_child_weight': 5.564497074247445, 'reg__feature_fraction': 0.8184851458696745, 'reg__bagging_fraction': 0.5657498212156961, 'reg__bagging_freq': 1, 'reg__max_bin': 103, 'reg__min_data_per_group': 127}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 7, 'reg__learning_rate': 0.02090586548861987, 'reg__num_leaves': 35, 'reg__subsample': 0.7112249596478034, 'reg__colsample_bytree': 0.6627527385443372, 'reg__min_child_samples': 44, 'reg__reg_alpha': 1.312650390096849, 'reg__reg_lambda': 26.315660819563156, 'reg__min_split_gain': 1.9643954206239111, 'reg__path_smooth': 1.7017977764548764, 'reg__min_child_weight': 1.45809519728512, 'reg__feature_fraction': 0.6506986596273165, 'reg__bagging_fraction': 0.5228171548943993, 'reg__bagging_freq': 5, 'reg__max_bin': 126, 'reg__min_data_per_group': 149}

3.5.2 Scatter



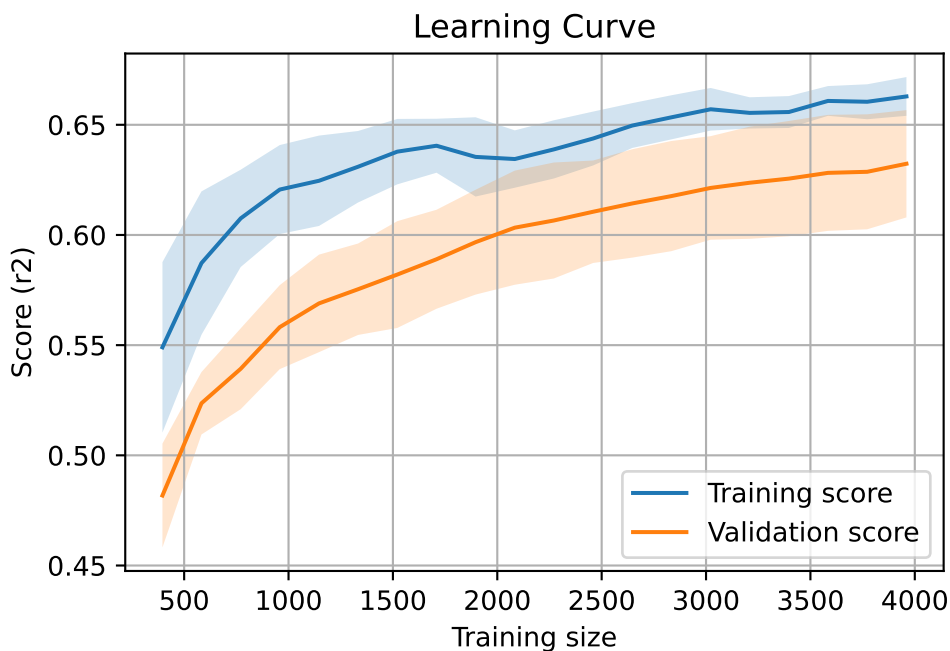
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 8084, `Len(training)` : 5659,

`Len(training_real)` : 3962, `Len(test_of_training)` : 1697, `Len(test)` : 2425



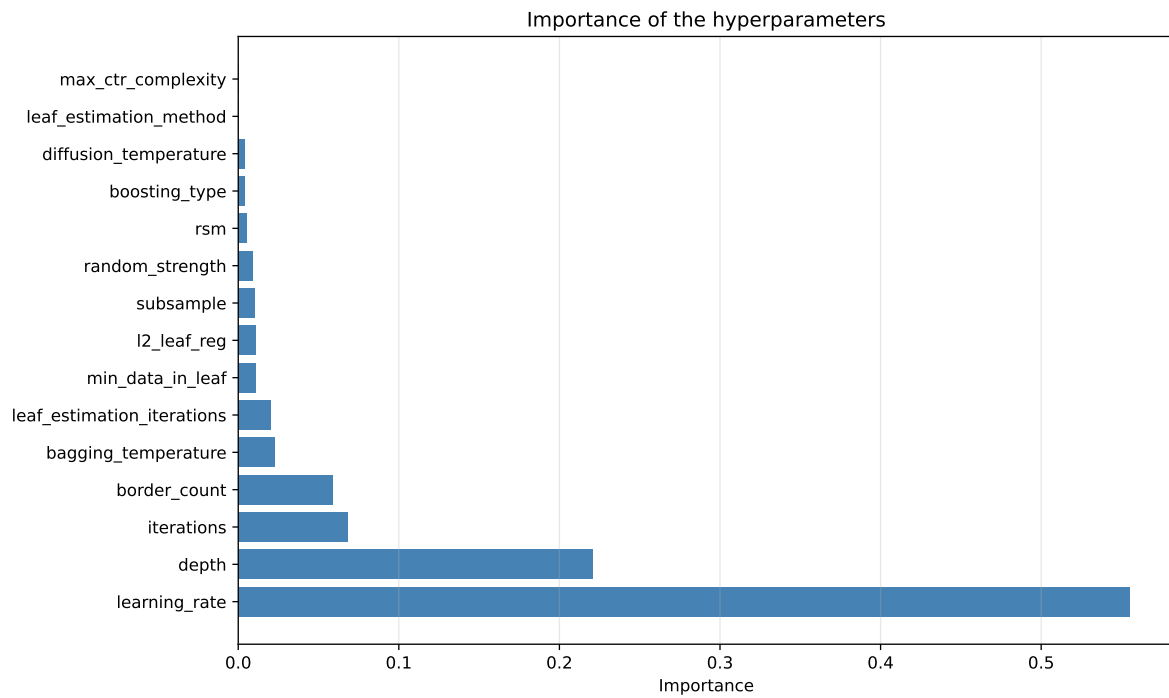
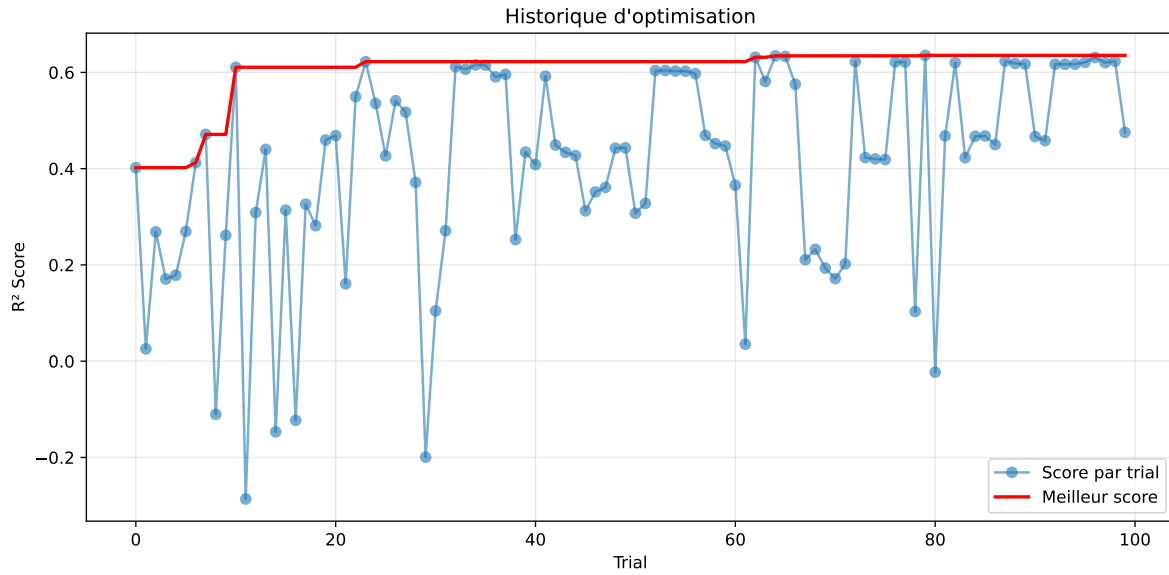
3.6 Model : CatBoost

3.6.1 Gridsearch

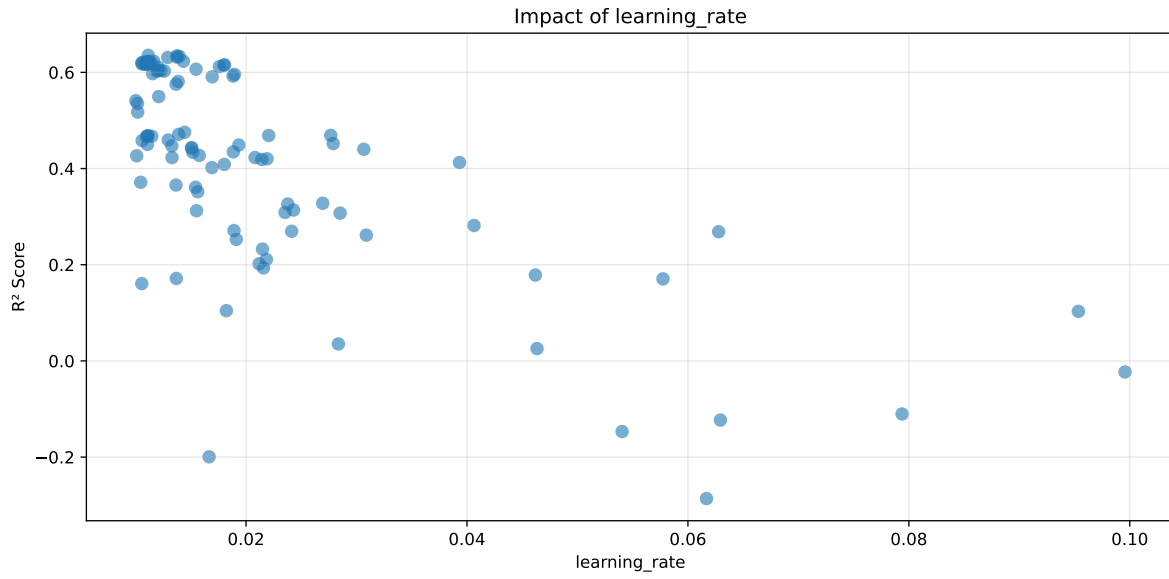
Distribution y_train vs y_test: y_train: mean=2.346, std=0.796, min=0.959, max=13.845
y_test: mean=2.362, std=0.797, min=0.973, max=7.779

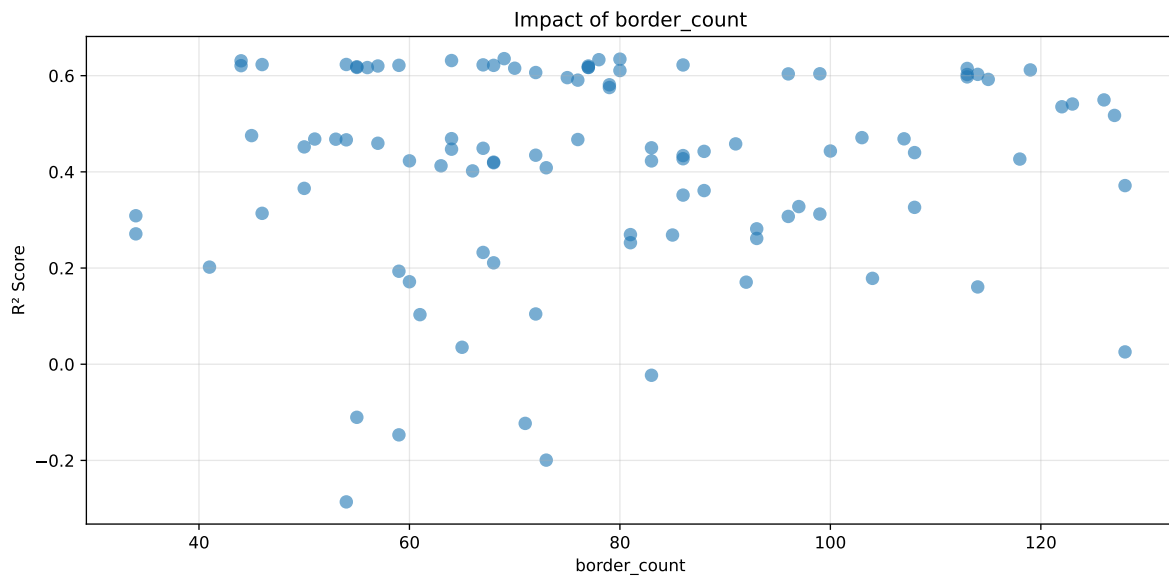
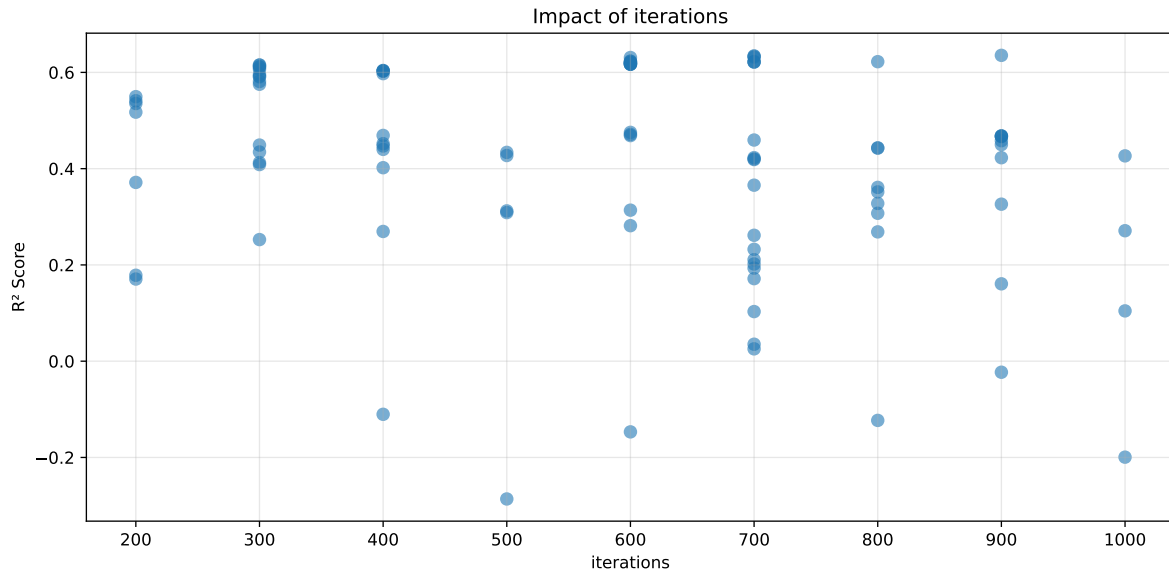
===== TOP Importance FEATURES =====

feature	importance
r_temperature_st_12h_slope	32.708289
r_umidity_st_12h_max	24.710431
r_temperature_st_12h_std	9.282790
r_CO2	8.718357
r_temperature_st_12h_diff_last	5.253810
r_NH3_st_12h_std	4.917300
r_umidity_st_12h_std	4.863918
r_NH3_st_12h_rms	4.663623
r_NH3_st_12h_slope	1.913570
r_umidity_st_12h_diff_last	1.732881
r_NH3_st_12h_diff_last	1.235031



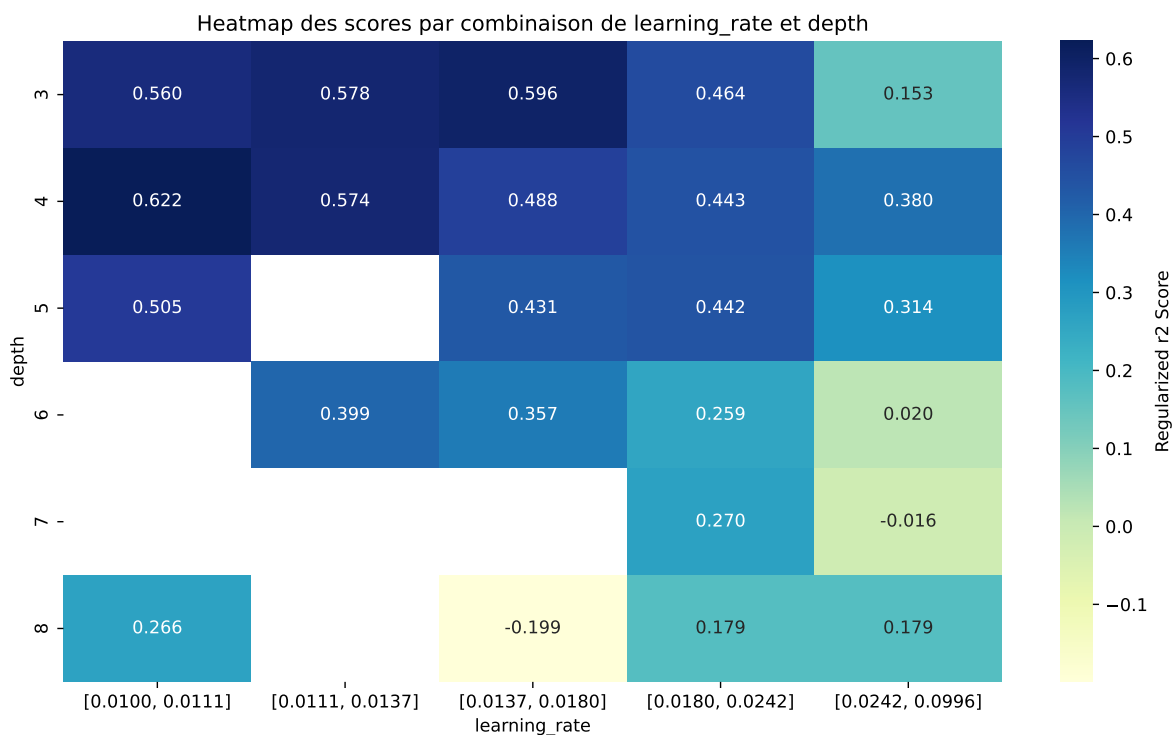
Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'iterations', 'border_count']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.667	0.6354	0.6242	0.0225	0.6354	0.6286	-0.0068	0.9893	1.0497
2	0.6648	0.6344	0.6228	0.0226	0.6344	0.6253	-0.009	0.9858	1.048
3	0.6637	0.6332	0.6216	0.0218	0.6332	0.626	-0.0072	0.9886	1.048
4	0.6622	0.6315	0.6202	0.022	0.6315	0.6236	-0.0079	0.9875	1.0487
5	0.661	0.6309	0.6174	0.0228	0.6309	0.6206	-0.0103	0.9837	1.0476
6	0.651	0.6233	0.6095	0.0214	0.6233	0.6127	-0.0105	0.9831	1.0445
7	0.6501	0.6231	0.6105	0.0213	0.6231	0.6141	-0.009	0.9855	1.0433
8	0.6485	0.6225	0.6095	0.0222	0.6225	0.6128	-0.0096	0.9845	1.0418
9	0.6492	0.6223	0.6092	0.0225	0.6223	0.6149	-0.0074	0.9881	1.0433
10	0.6483	0.6215	0.6092	0.0223	0.6215	0.6107	-0.0108	0.9826	1.0432
---	---	---	---	---	---	---	---	---	---
100	0.9359	0.7322	0.7319	0.0221	-0.2863	0.7406	0.0085	1.0115	1.2782

Hyperparameters:

Rank 1: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.011140818047081776, 'reg__l2_leaf_reg': 44.163485272537606, 'reg__bagging_temperature': 2.049613181235965, 'reg__random_strength': 2.0461774826635155, 'reg__subsample': 0.5093382570678713, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.5812291478595196, 'reg__border_count': 69, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9868.104738709568}

Rank 2: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.013733818272850801, 'reg__l2_leaf_reg': 47.18984316755815, 'reg__bagging_temperature': 2.60017174012271, 'reg__random_strength': 2.4929362643873816, 'reg__subsample': 0.857648921525135, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.8537696604948514, 'reg__border_count': 80, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4470.687719904012}

Rank 3: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.013954791935795562, 'reg__l2_leaf_reg': 46.157151903293126, 'reg__bagging_temperature': 2.6376346788681113, 'reg__random_strength': 2.677857961918185, 'reg__subsample': 0.8555572541902645, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.8489916173669567, 'reg__border_count': 78, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 9875.104533750538}

Rank 4: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.01376418530003567, 'reg__l2_leaf_reg': 47.76006627864345, 'reg__bagging_temperature': 2.6105328236600145, 'reg__random_strength': 2.521617021310442, 'reg__subsample': 0.587143481459964, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.7082331153534628, 'reg__border_count': 64, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 4795.4354641157715}

Rank 5: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.012912711075489056, 'reg__l2_leaf_reg': 46.47442750979255, 'reg__bagging_temperature': 3.2463354896927856, 'reg__random_strength': 0.8817853784360625, 'reg__subsample': 0.5214786634050161, 'reg__min_data_in_leaf': 49, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.5502414385128824, 'reg__border_count': 44, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 9316.35566302263}

Rank 6: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.011635260749444308, 'reg__l2_leaf_reg': 44.21811259704928, 'reg__bagging_temperature': 3.02658708862492, 'reg__random_strength': 1.24479194235736, 'reg__subsample': 0.5148999414948191, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.602327275869737, 'reg__border_count': 54, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7410.186277349409}

Rank 7: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.014326060840177852, 'reg__l2_leaf_reg': 42.300186965479014, 'reg__bagging_temperature': 3.7181161448613844, 'reg__random_strength': 2.10984277119642, 'reg__subsample': 0.5220615081655915,

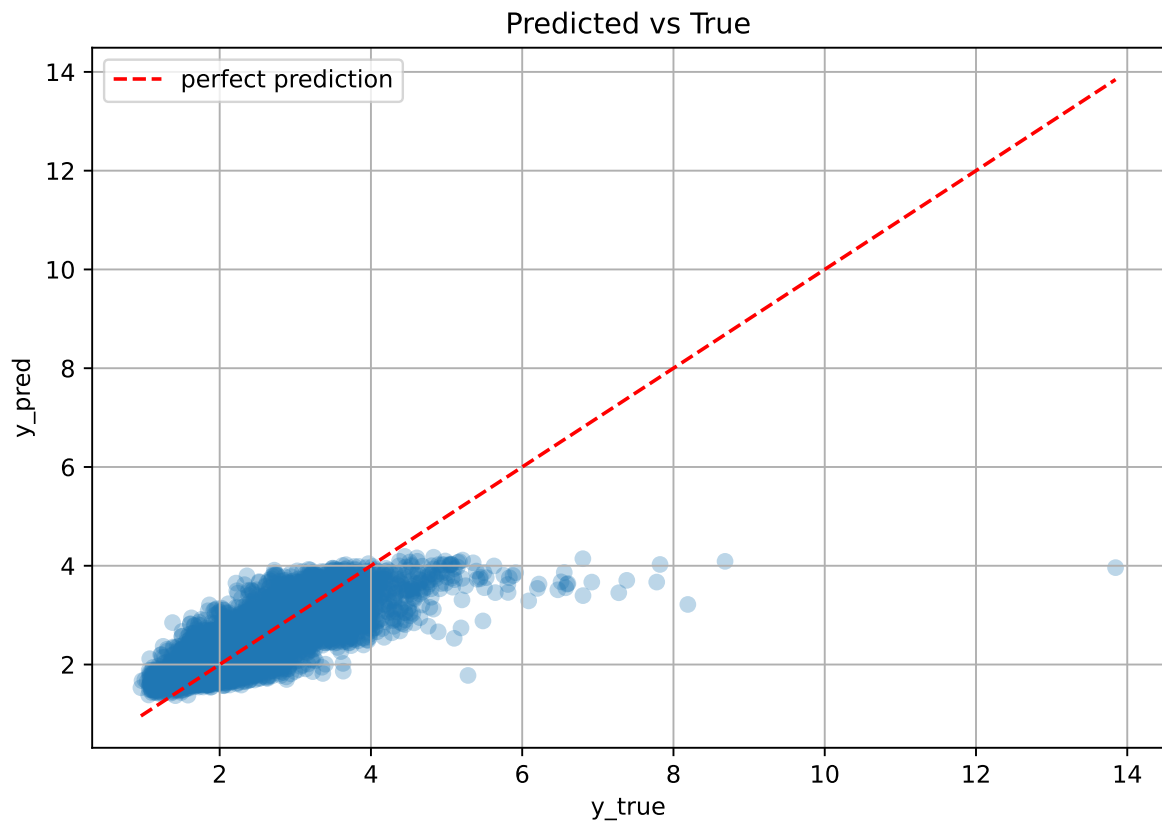
'reg__min_data_in_leaf': 23, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.5198040786754747, 'reg__border_count': 46, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4160.178534126648}

Rank 8: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.011190415720281743, 'reg__l2_leaf_reg': 48.30645219808853, 'reg__bagging_temperature': 3.2062028370701032, 'reg__random_strength': 2.005556091893152, 'reg__subsample': 0.5437789257424505, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.6149047558401902, 'reg__border_count': 67, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 5356.841535558124}

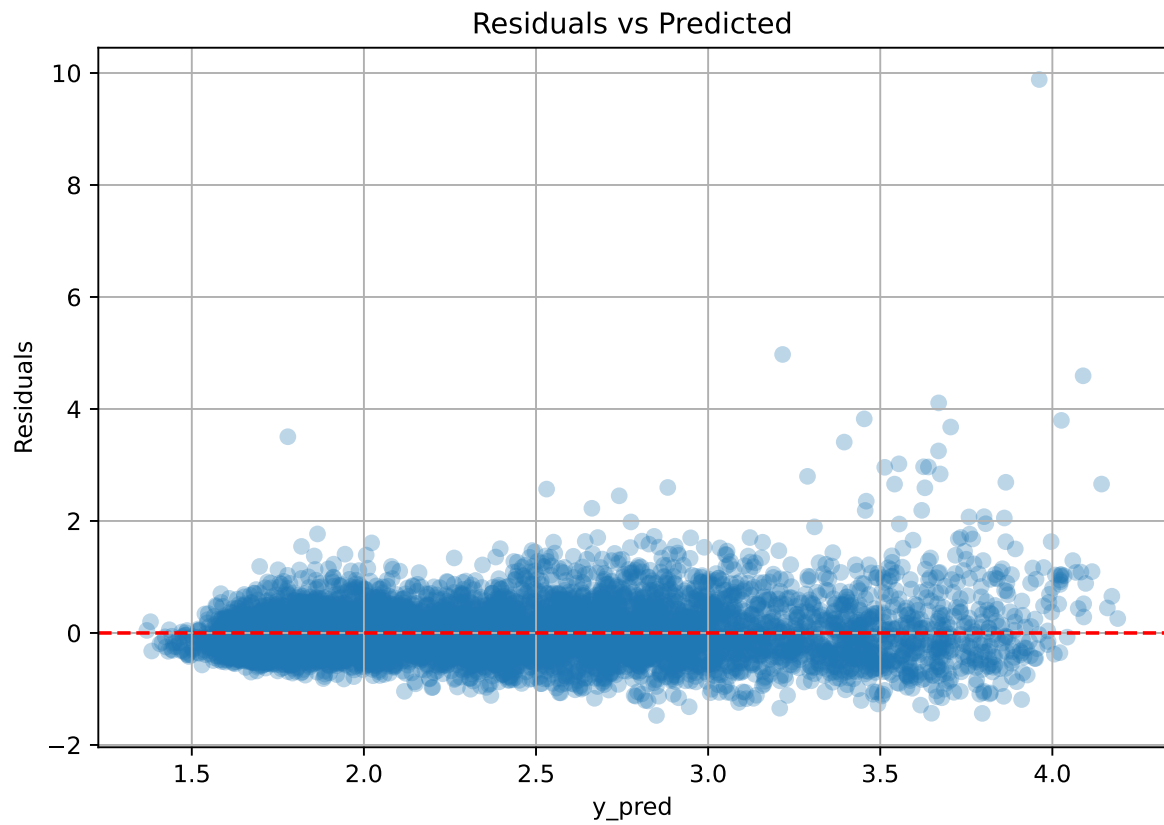
Rank 9: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.011037603129032923, 'reg__l2_leaf_reg': 18.62218096791285, 'reg__bagging_temperature': 0.9199942167392534, 'reg__random_strength': 3.6219768992713406, 'reg__subsample': 0.6430526582191293, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8654903784797661, 'reg__border_count': 86, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 425.81261731187897}

Rank 10: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.01103831090464478, 'reg__l2_leaf_reg': 48.03926742384398, 'reg__bagging_temperature': 3.2022063805718917, 'reg__random_strength': 1.8739991780995193, 'reg__subsample': 0.5323321563901671, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.5874426491241365, 'reg__border_count': 59, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5597.278713446606}

3.6.2 Scatter



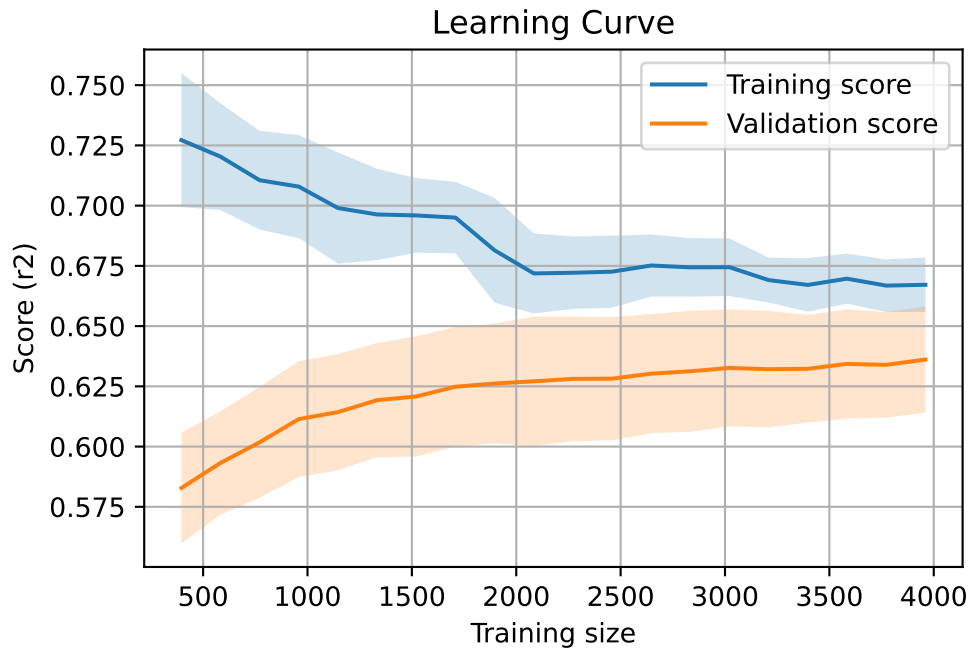
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 8084, `Len(training)` : 5659,

`Len(training_real)` : 3962, `Len(test_of_training)` : 1697, `Len(test)` : 2425



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.529	0.521	0.02	0.503	0.503	-0.018	0.966	1.015	nan
LGBM	0.662	0.632	0.024	0.62	0.638	0.006	1.009	1.049	0.6318
CatBoost	0.667	0.635	0.022	0.624	0.629	-0.007	0.989	1.05	0.6354

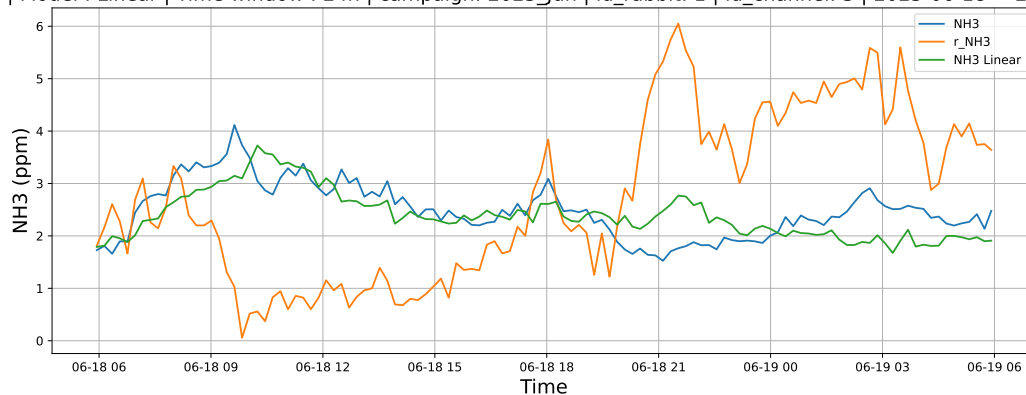
4 Plot

4.1 Standalone plots

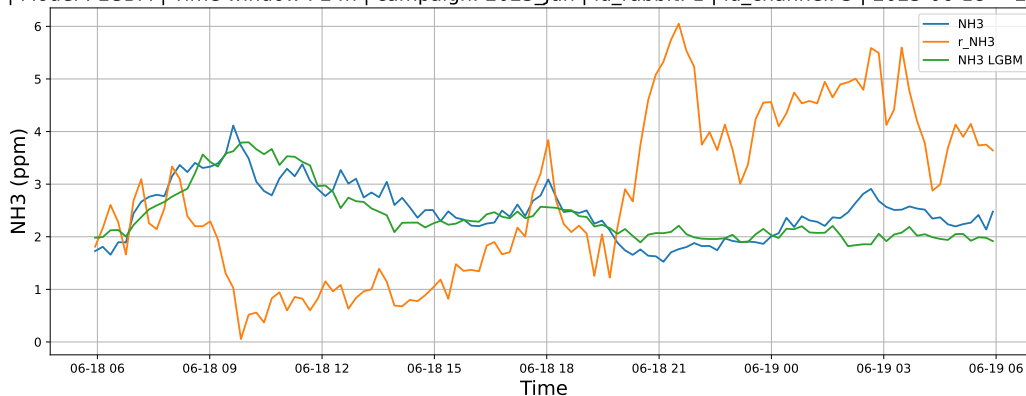
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

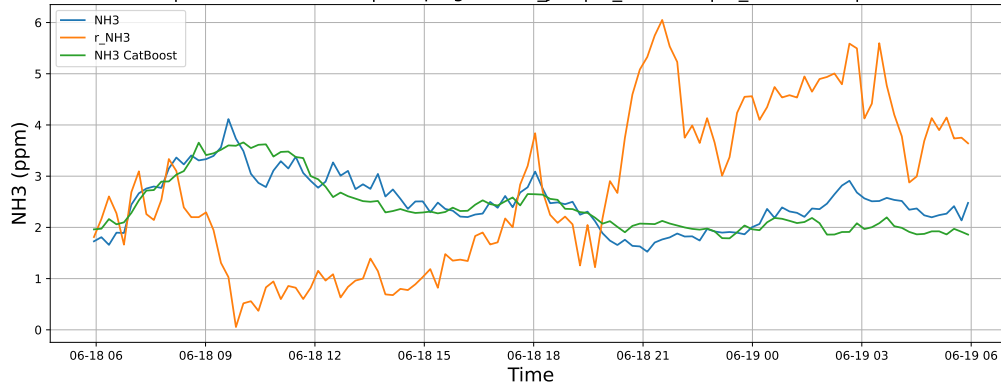
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

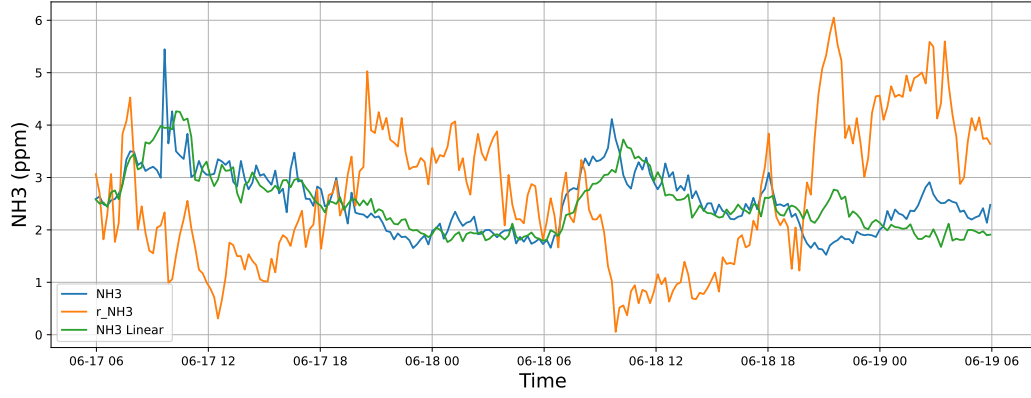


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

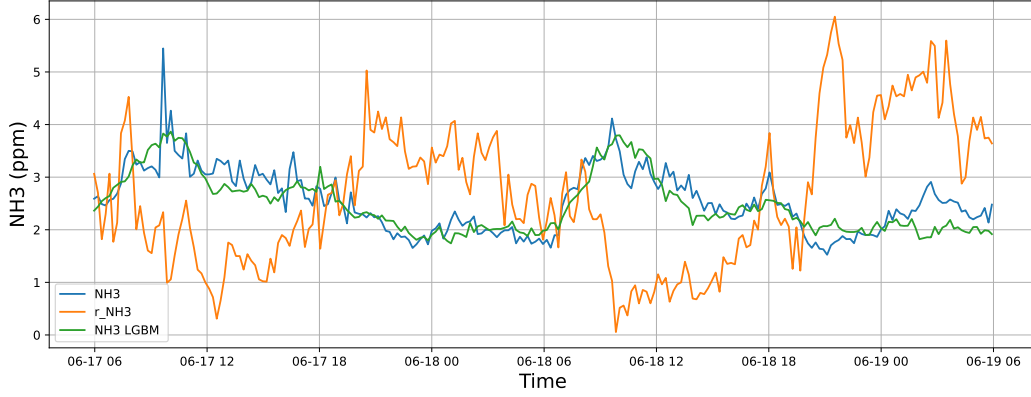


4.1.1.2 Time window : 48

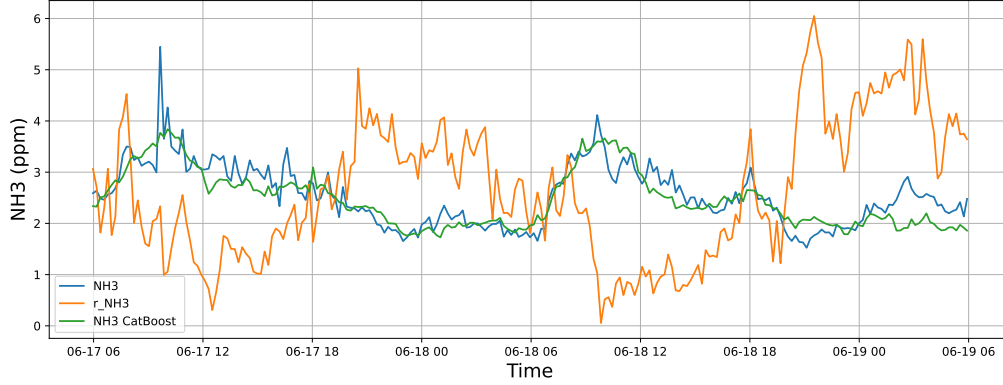
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

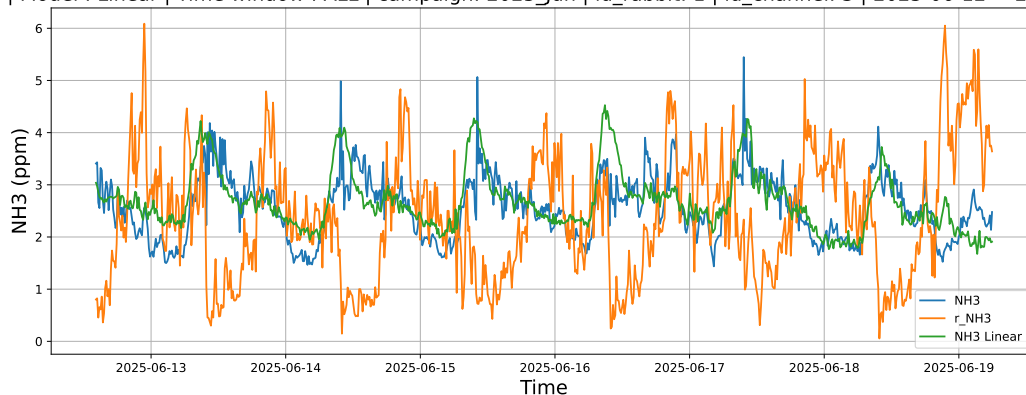


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

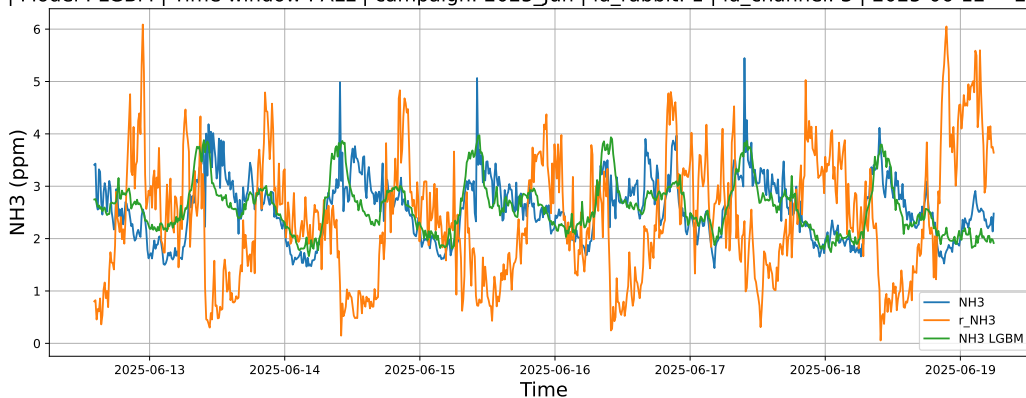


4.1.1.3 Time window : ALL

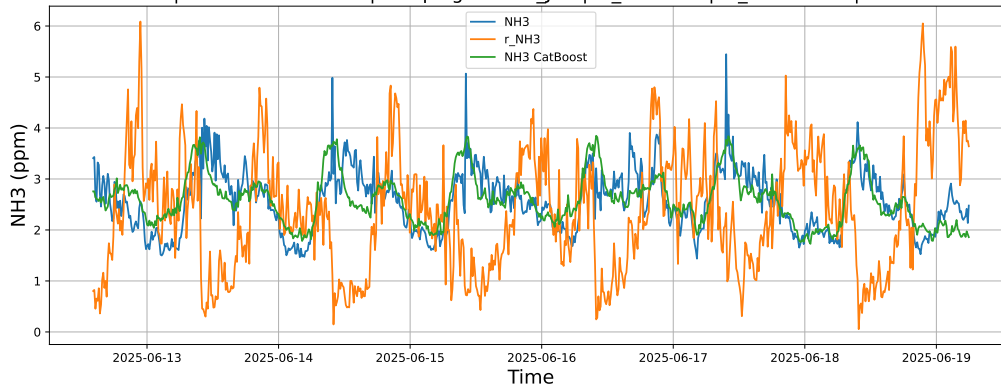
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



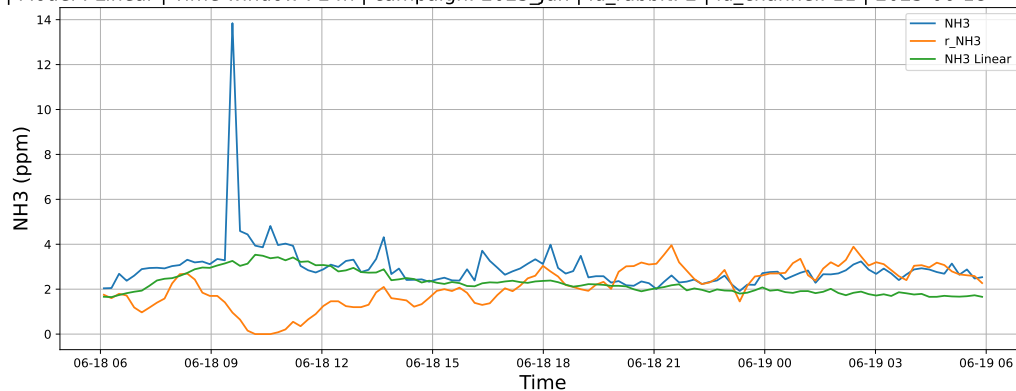
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



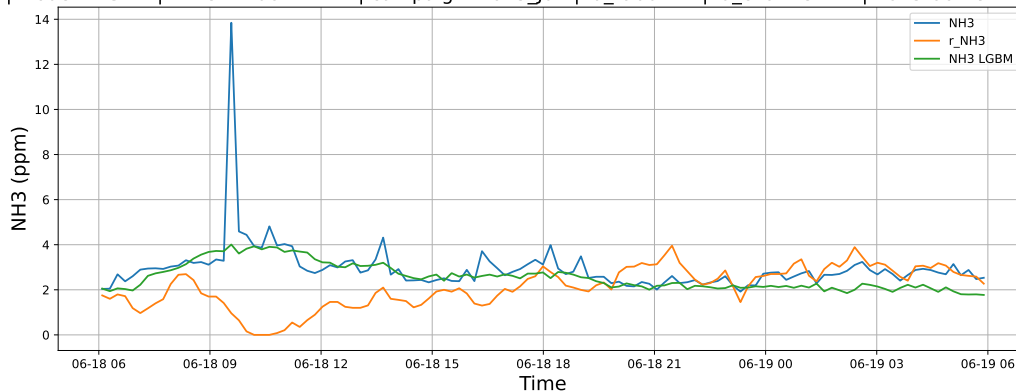
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

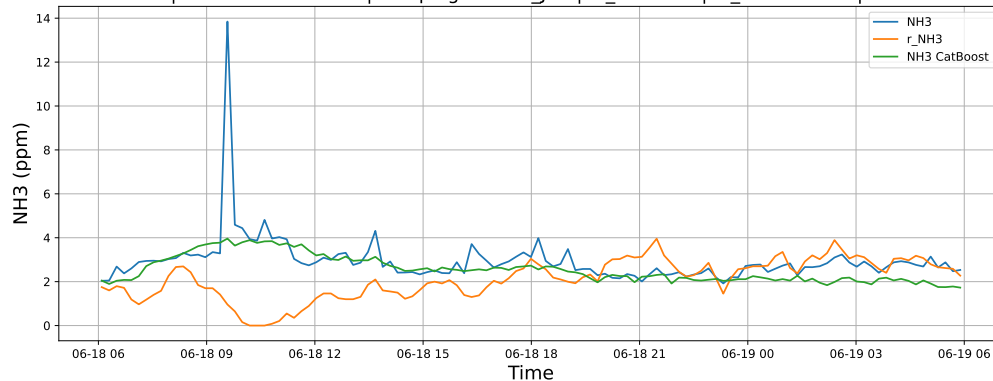
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

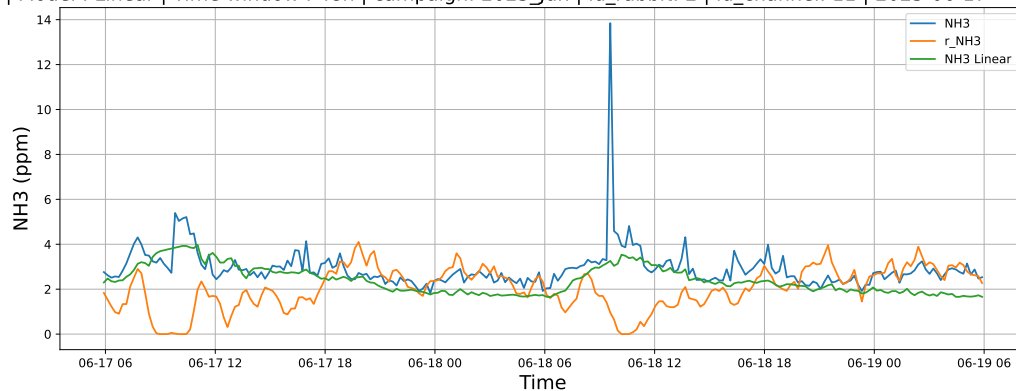


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

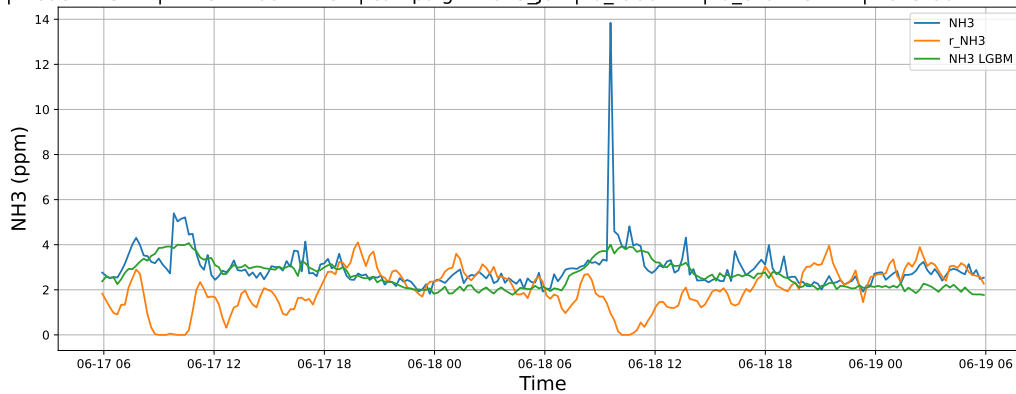


4.1.2.2 Time window : 48

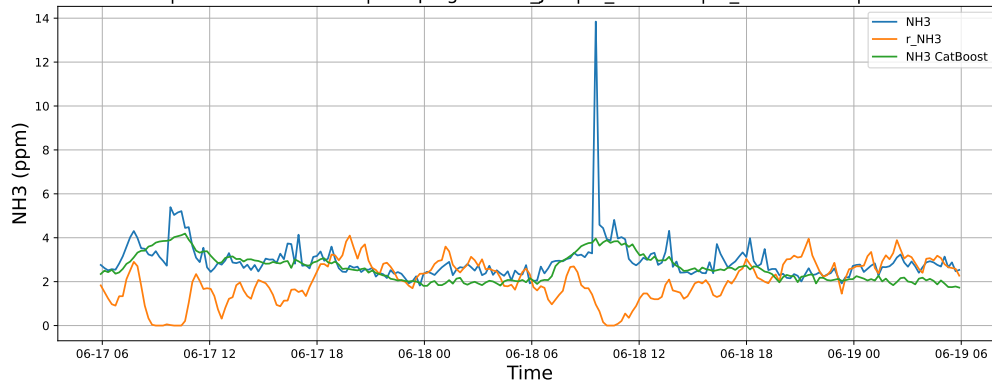
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

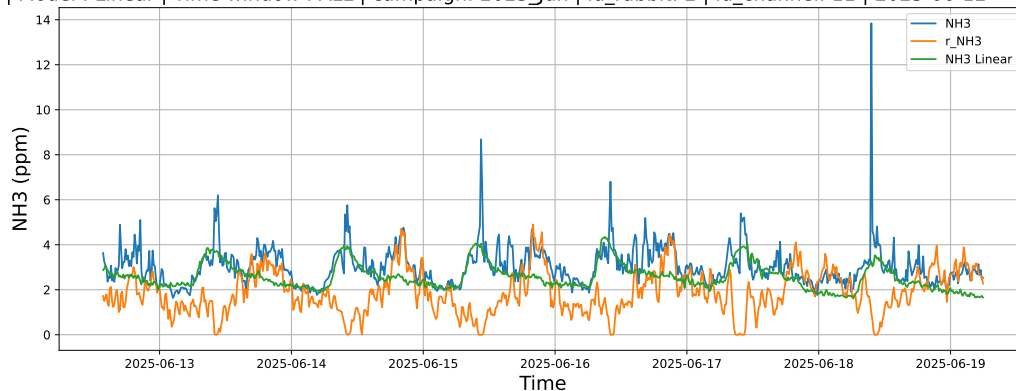


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

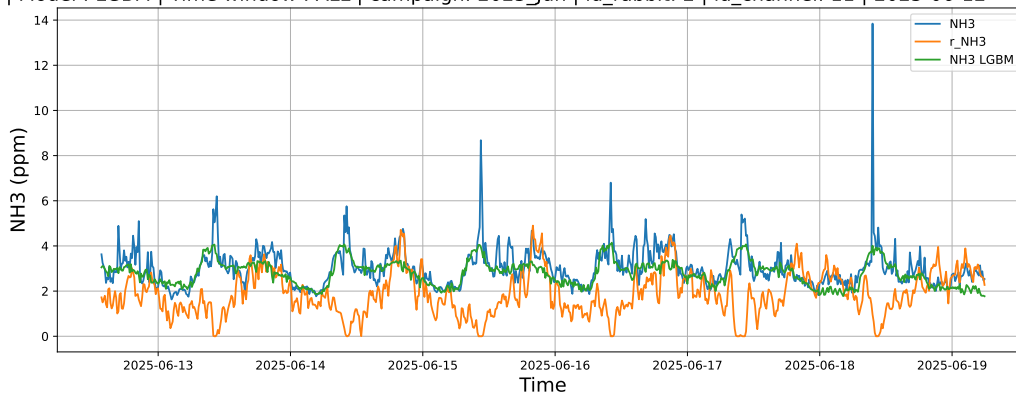


4.1.2.3 Time window : ALL

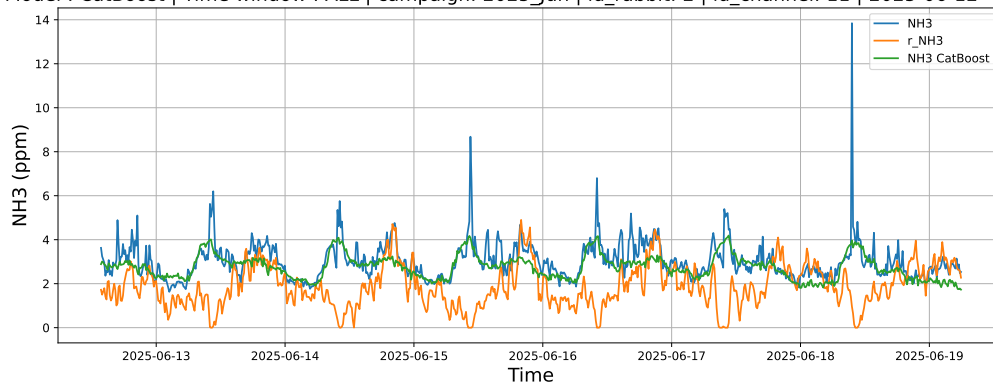
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



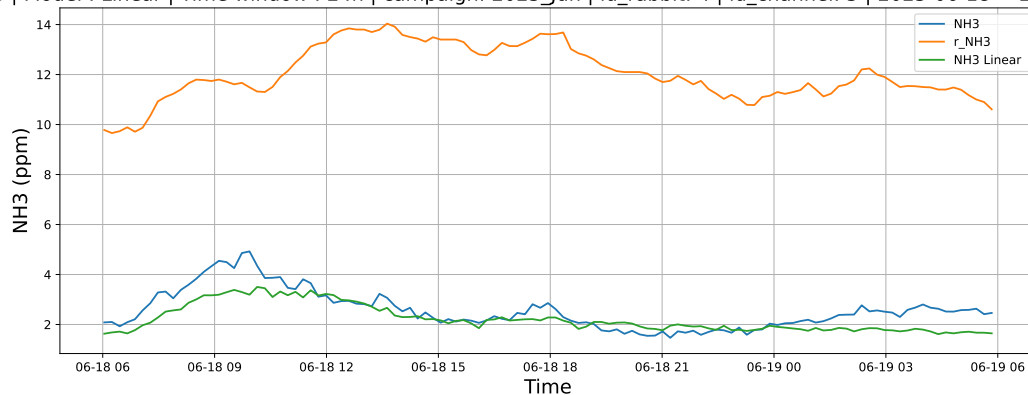
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



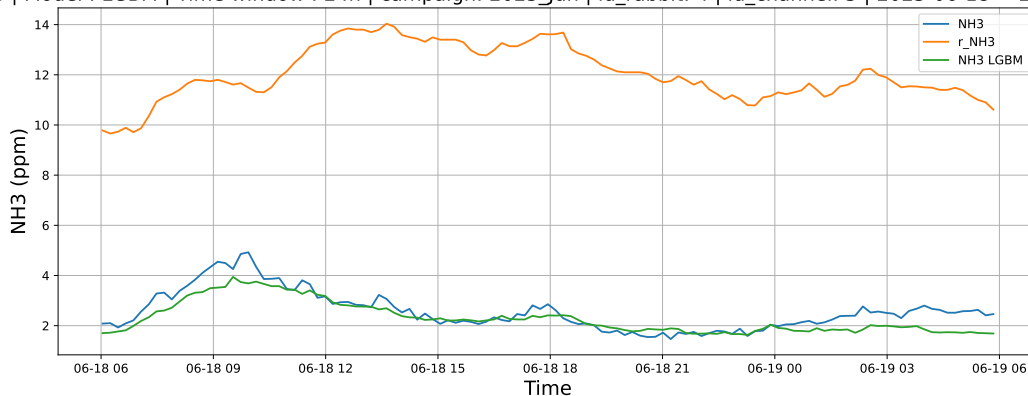
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

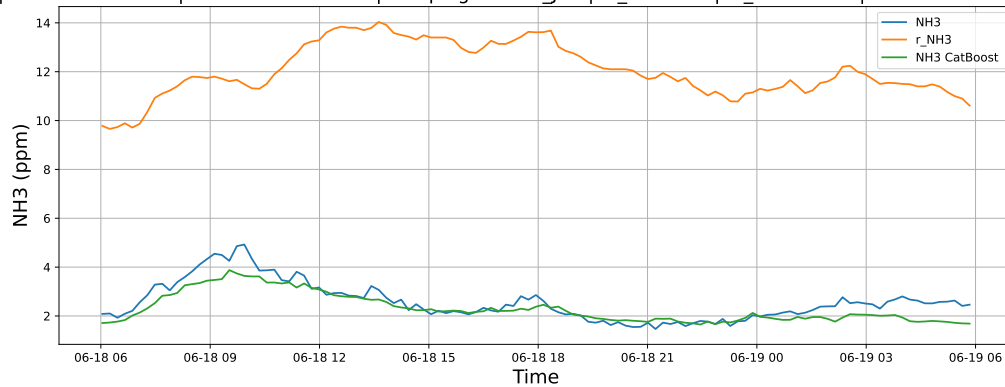
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

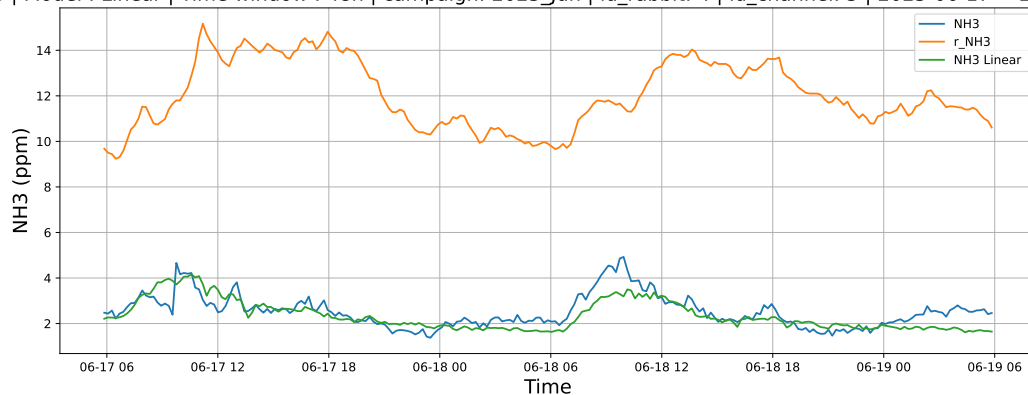


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

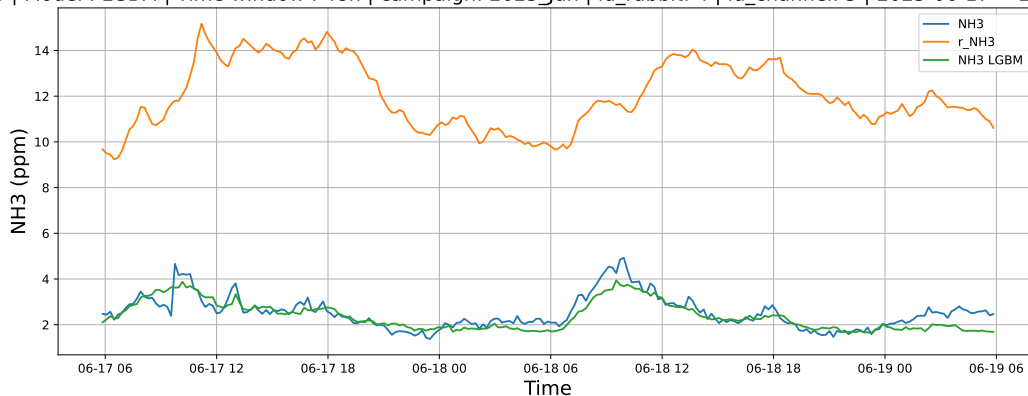


4.1.3.2 Time window : 48

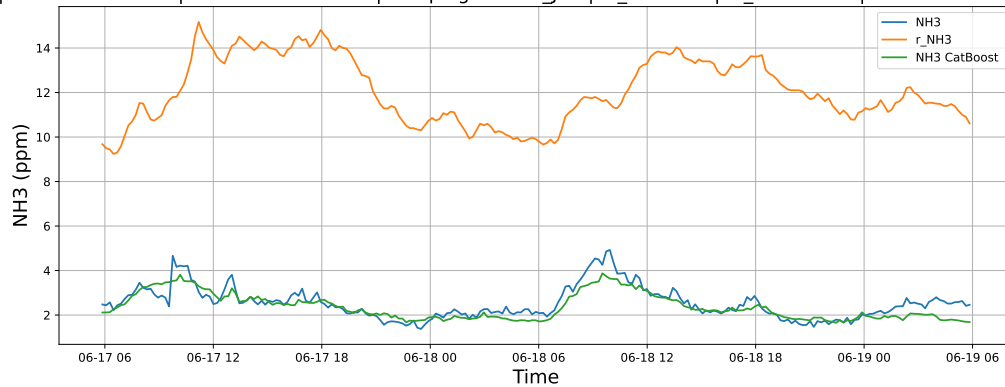
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

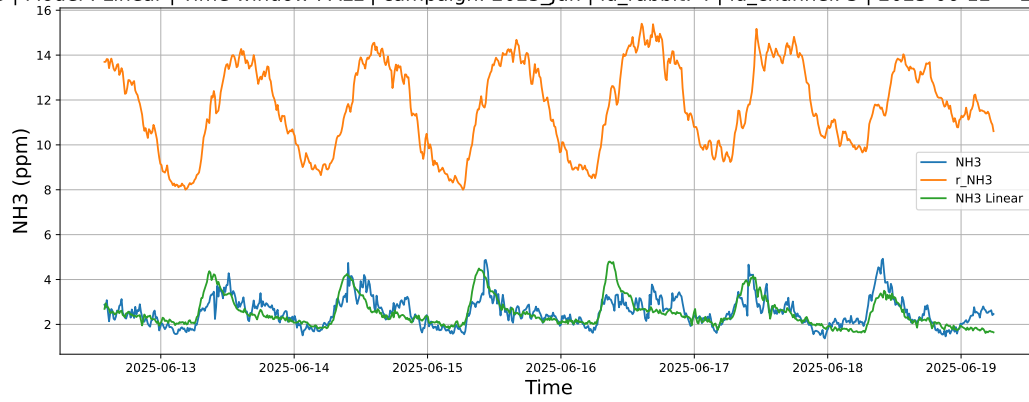


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

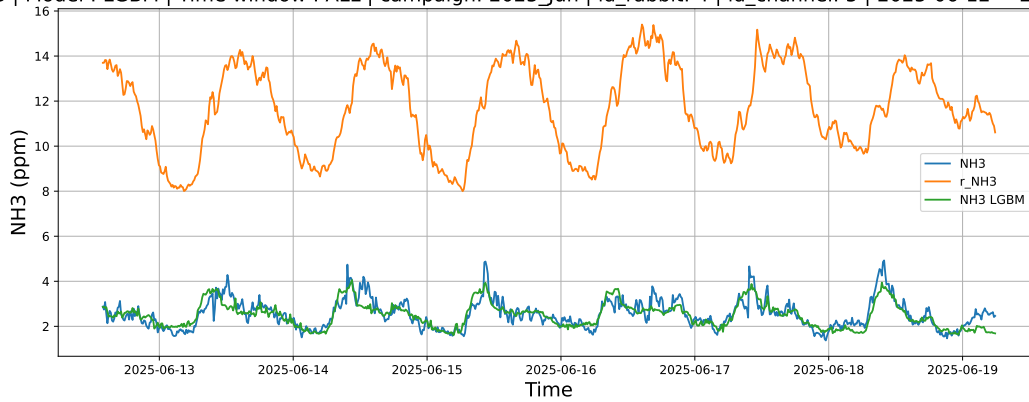


4.1.3.3 Time window : ALL

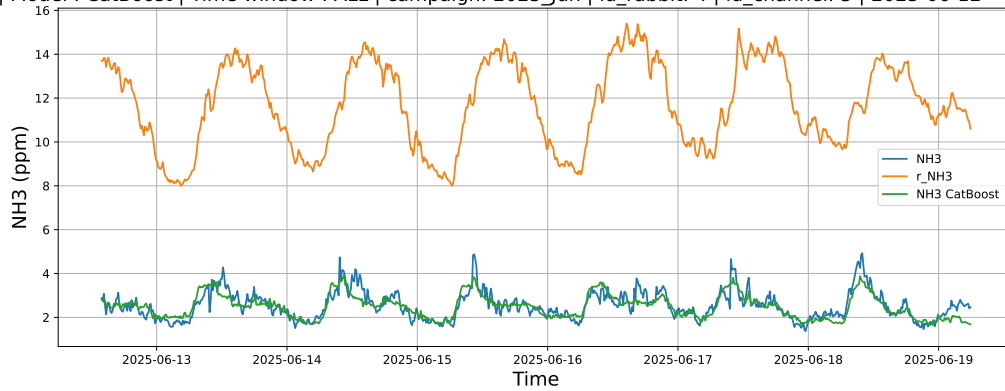
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



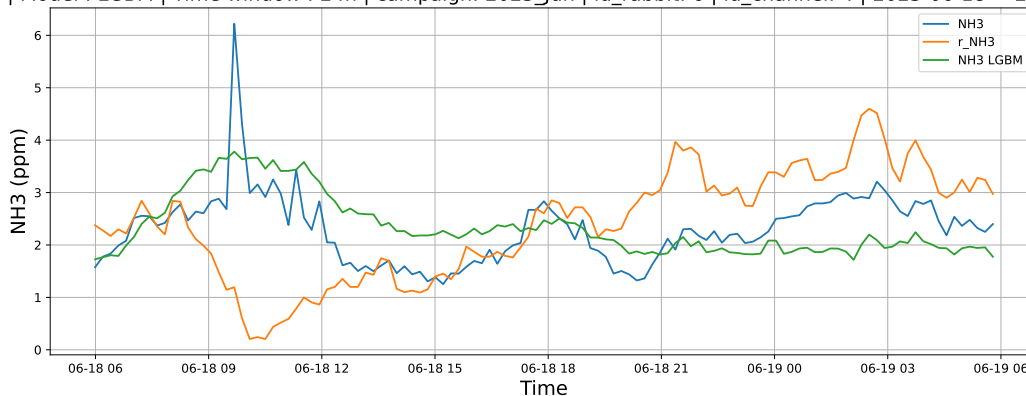
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

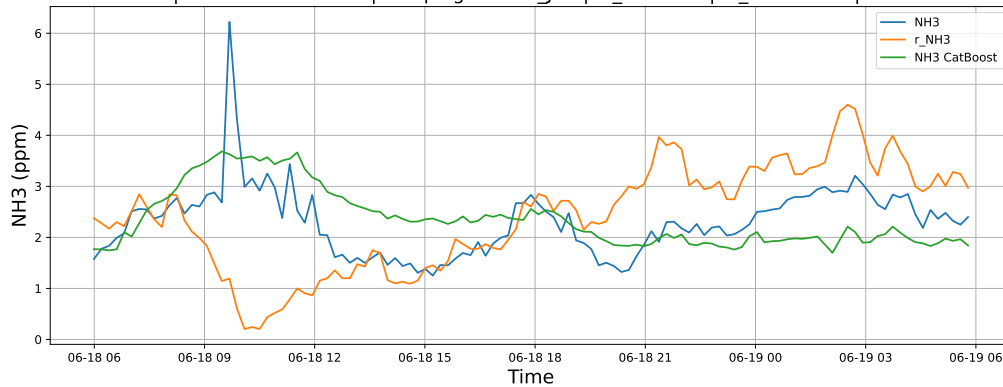
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

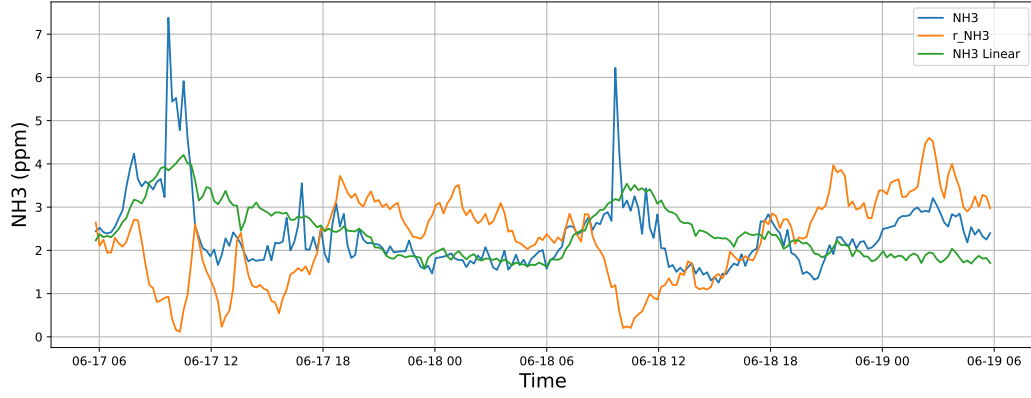


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

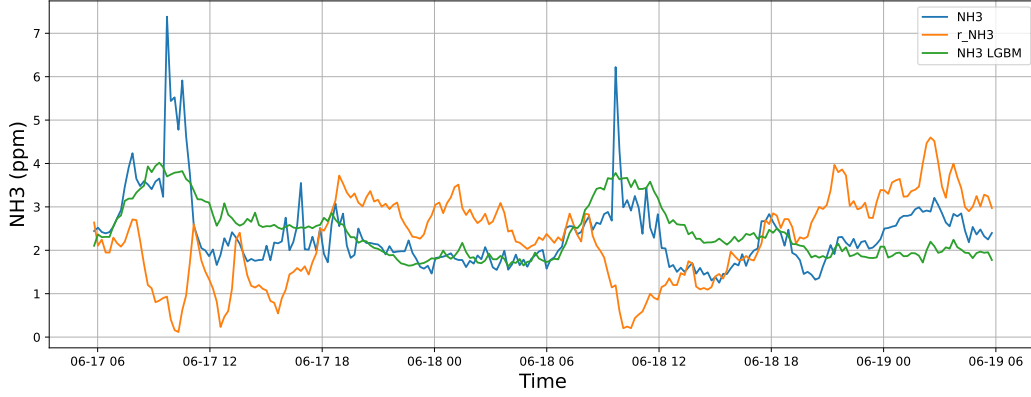


4.1.4.2 Time window : 48

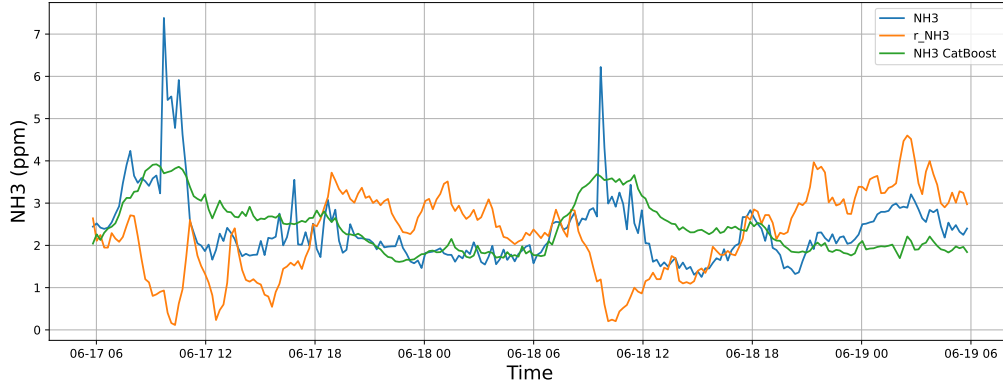
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

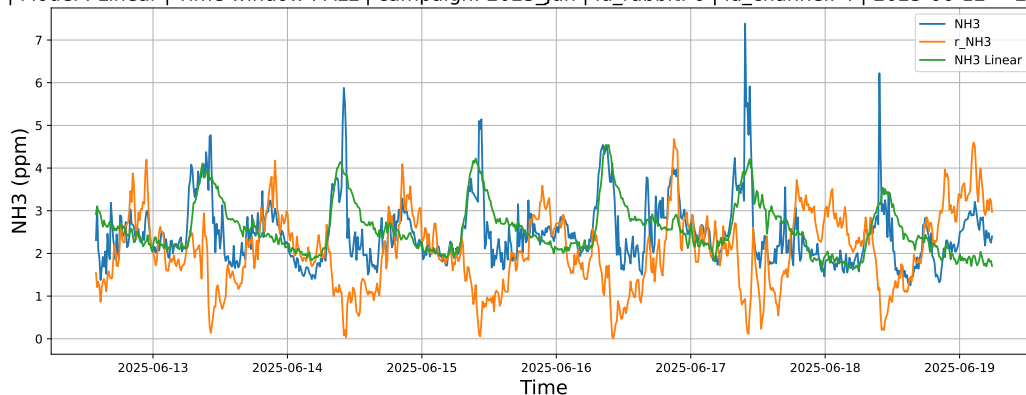


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

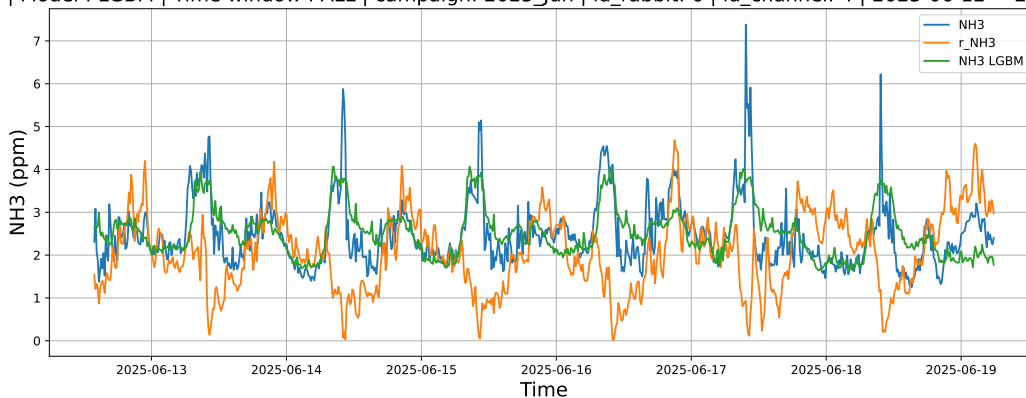


4.1.4.3 Time window : ALL

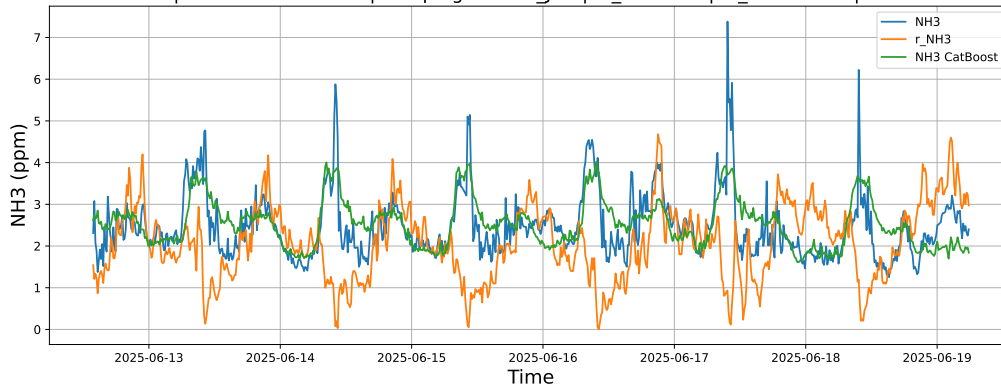
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



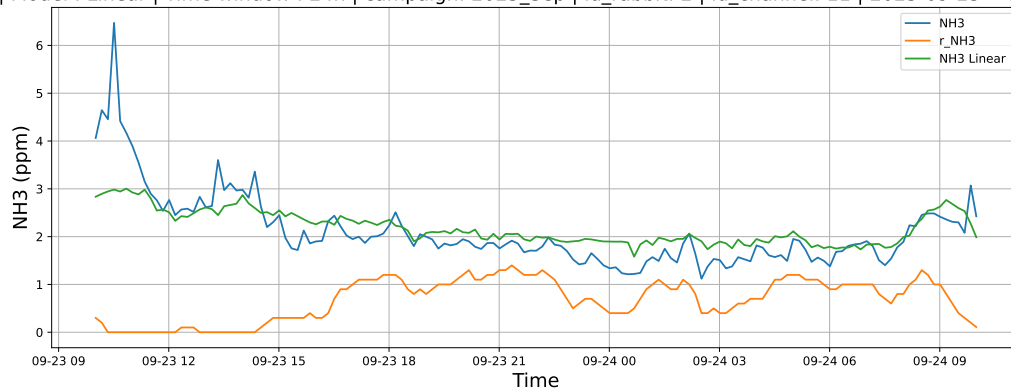
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



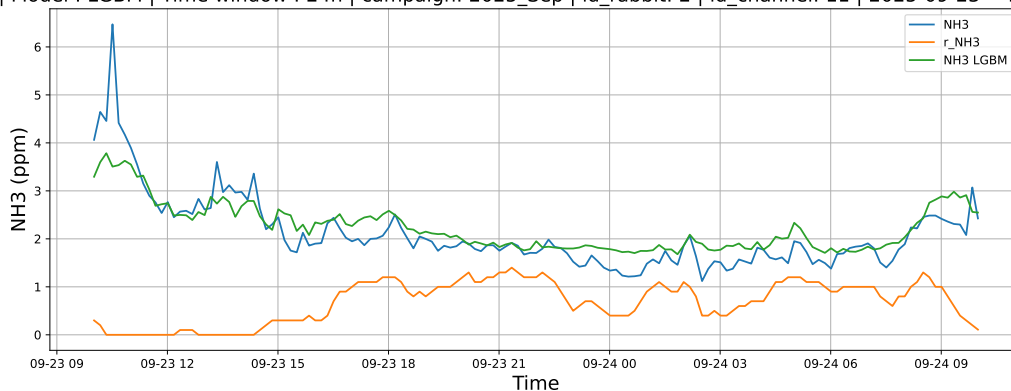
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

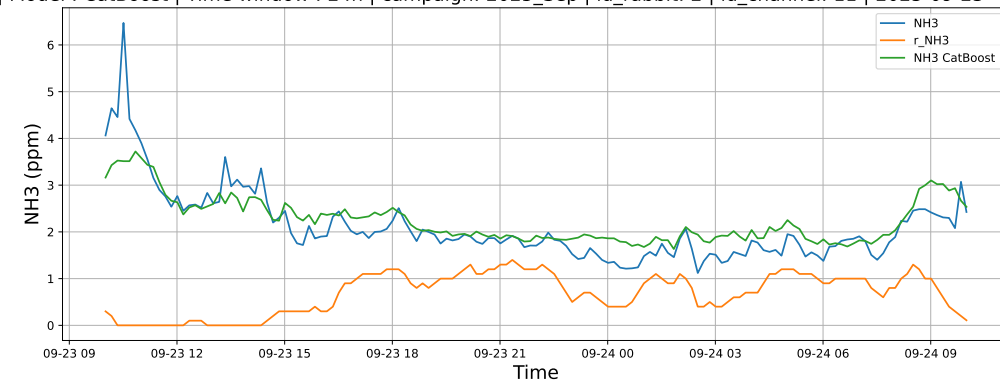
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

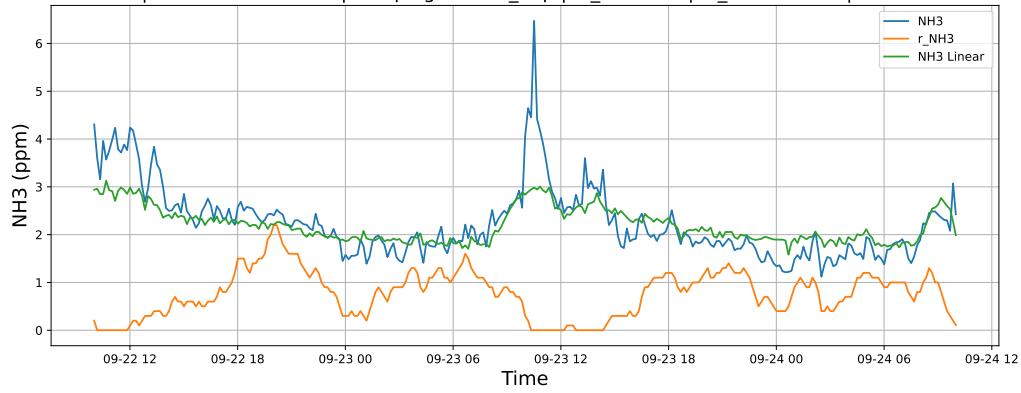


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

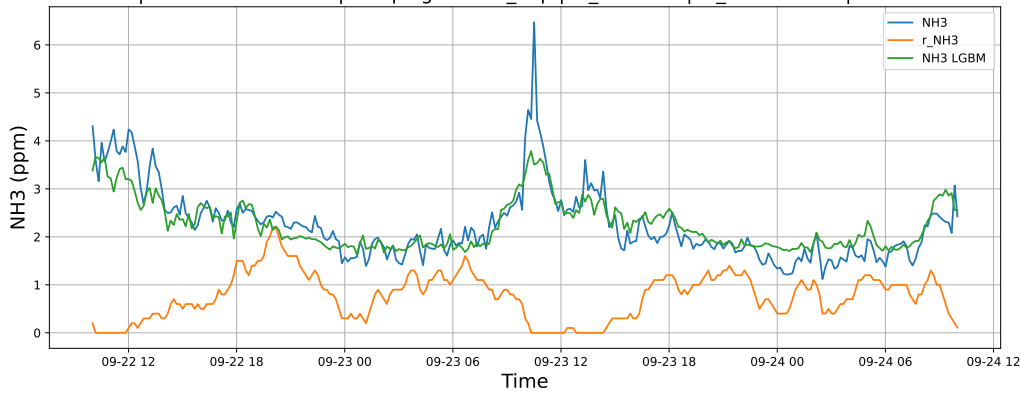


4.1.5.2 Time window : 48

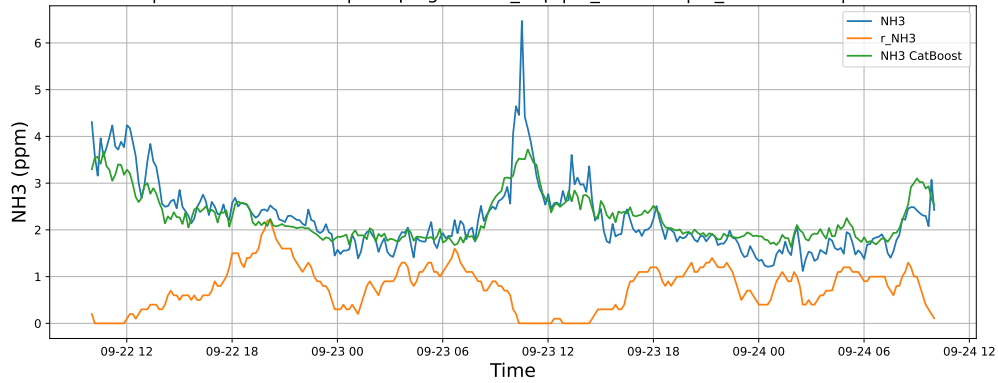
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

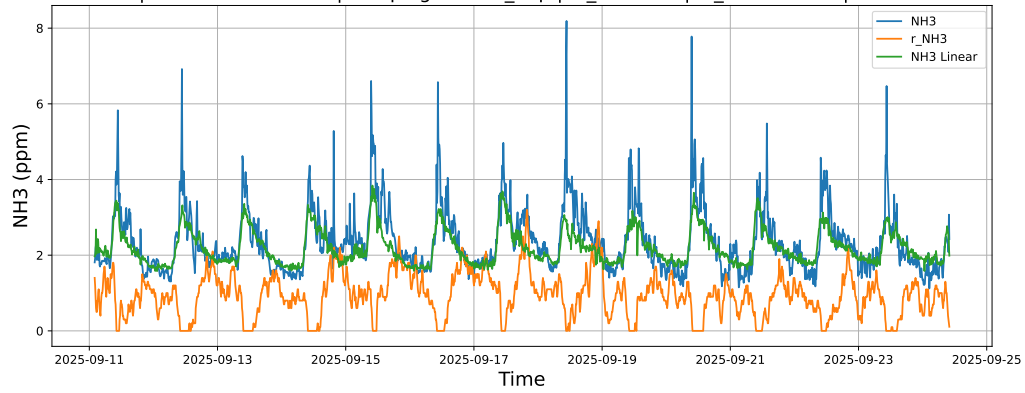


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

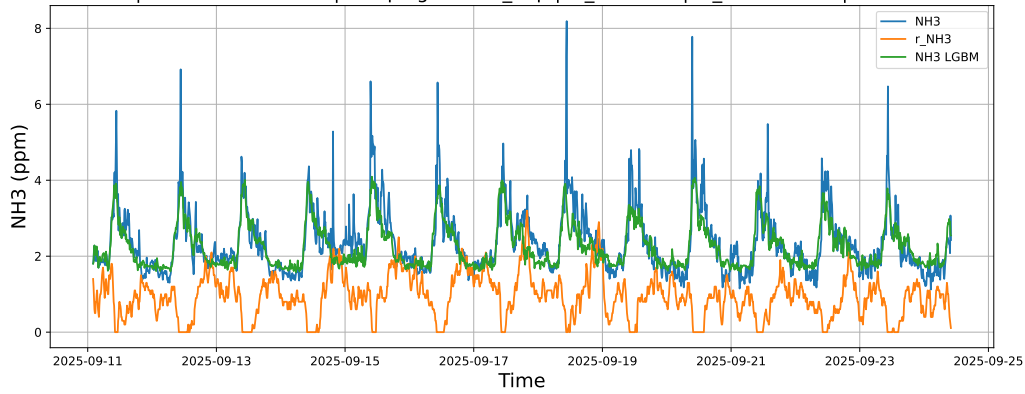


4.1.5.3 Time window : ALL

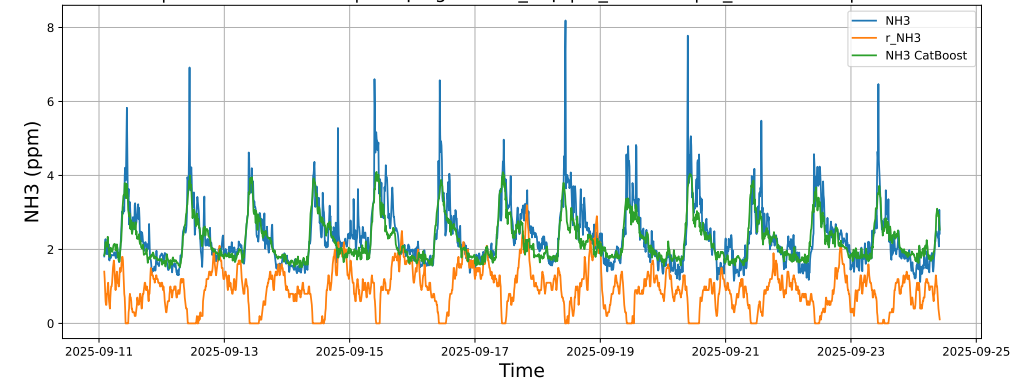
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



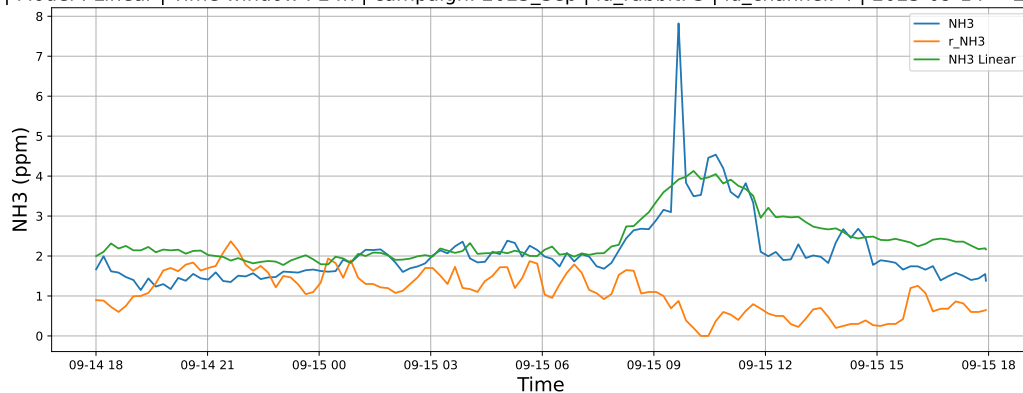
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



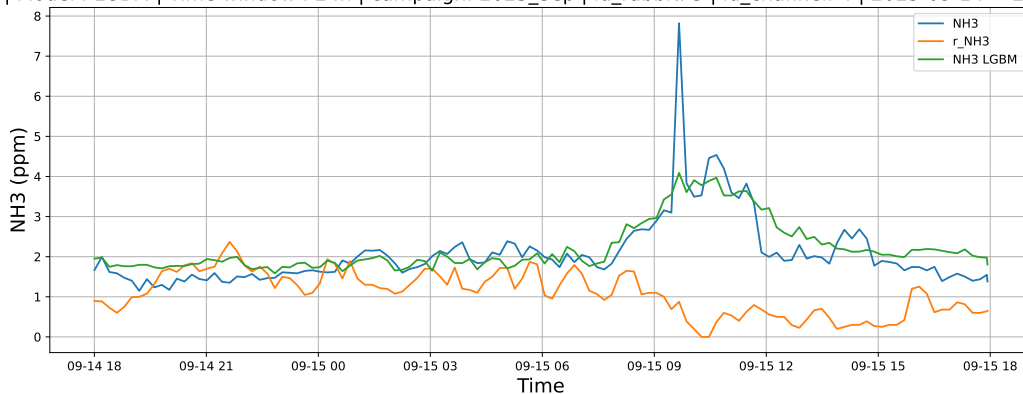
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

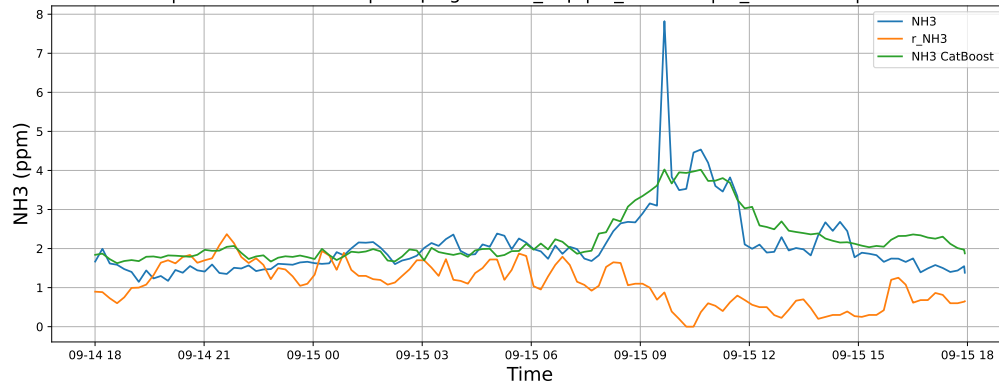
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

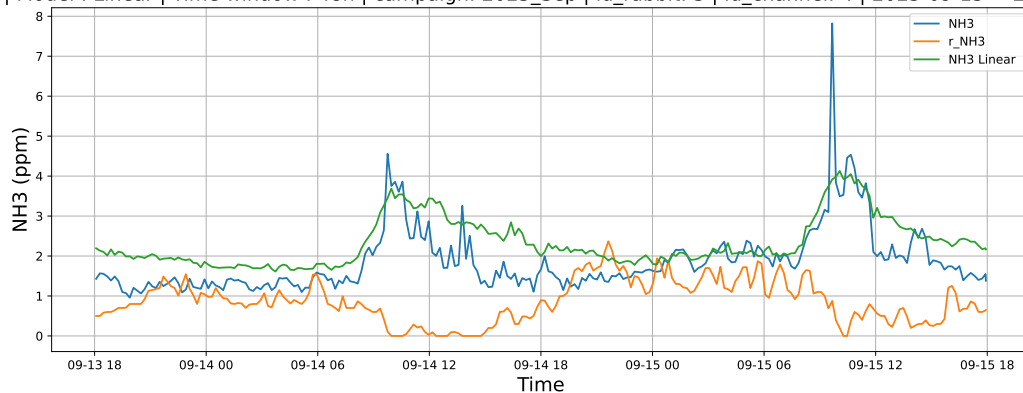


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

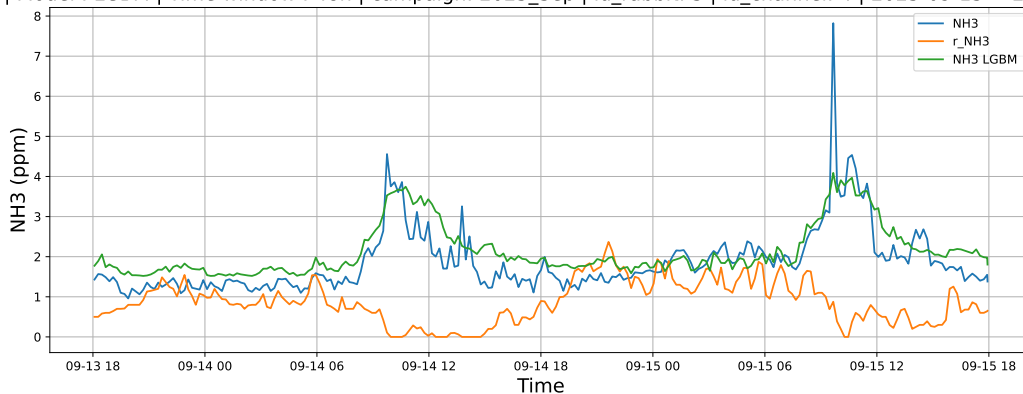


4.1.6.2 Time window : 48

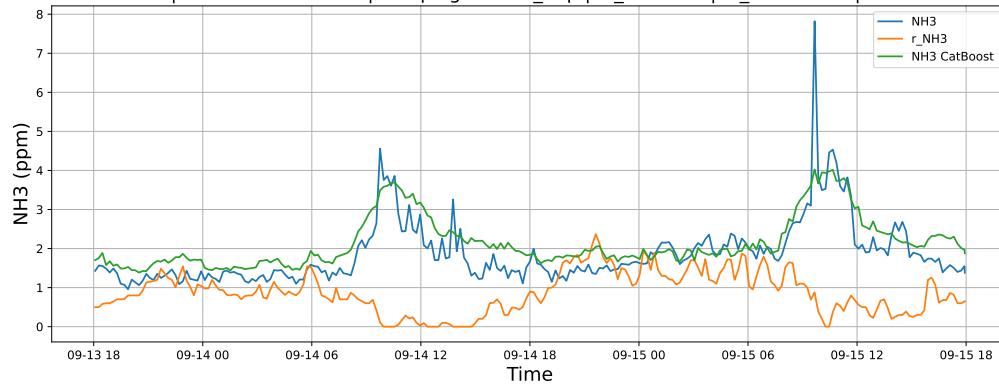
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

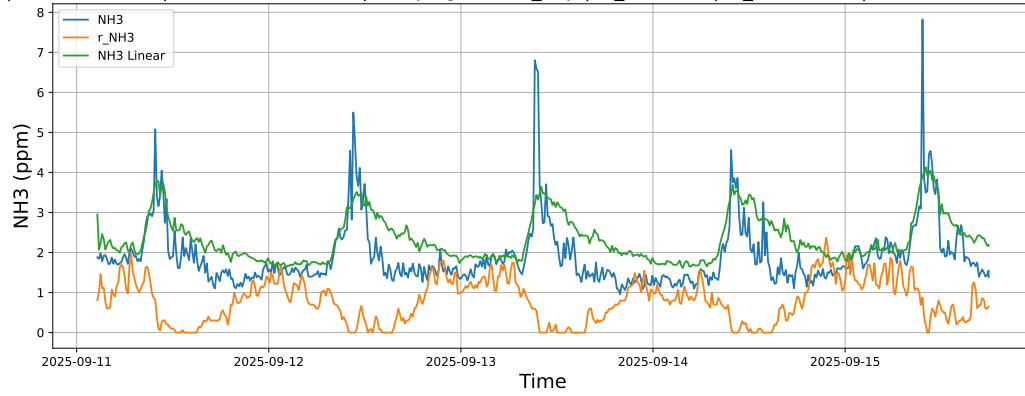


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

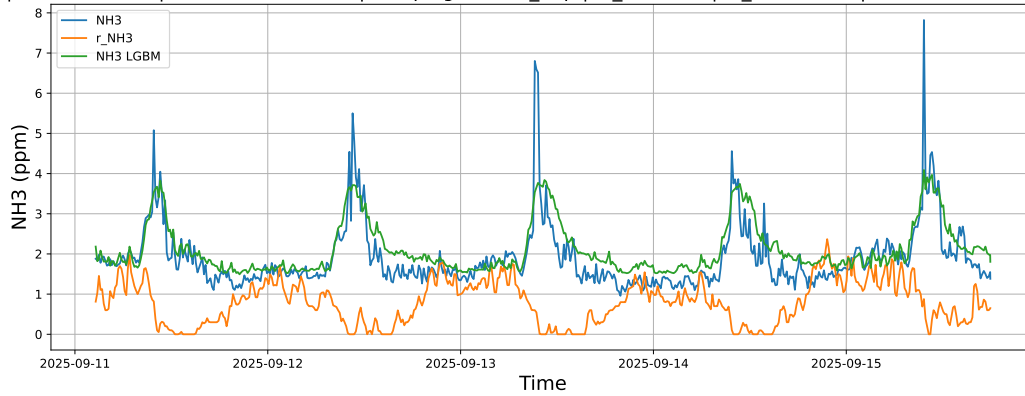


4.1.6.3 Time window : ALL

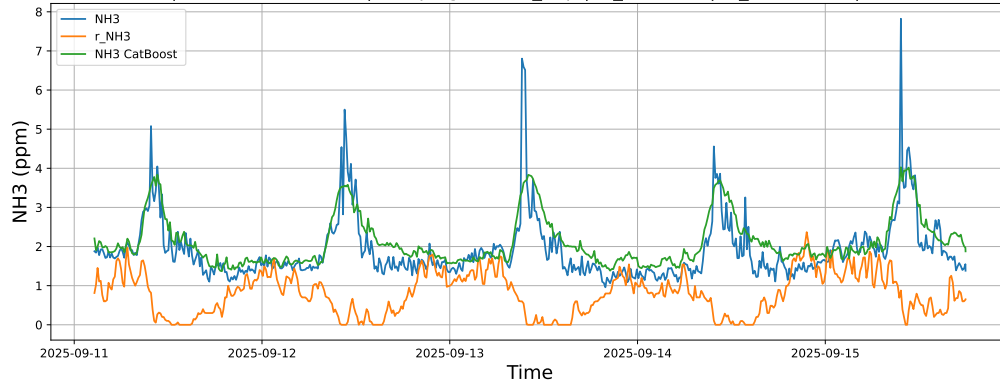
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



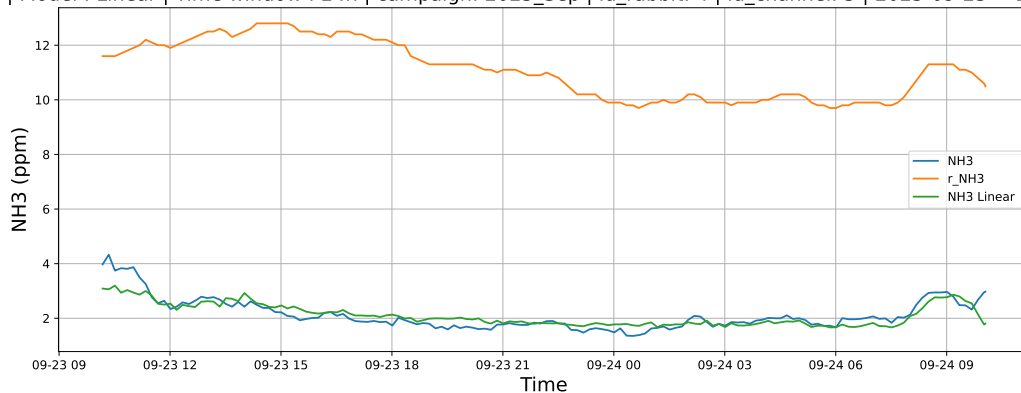
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



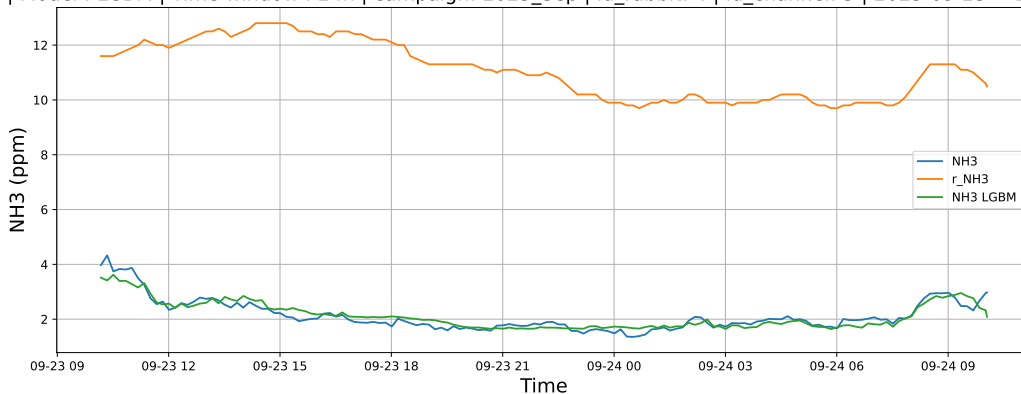
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

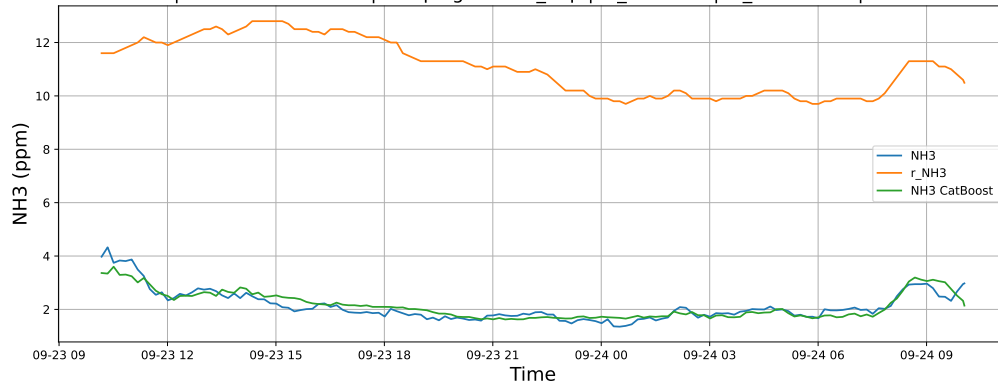
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

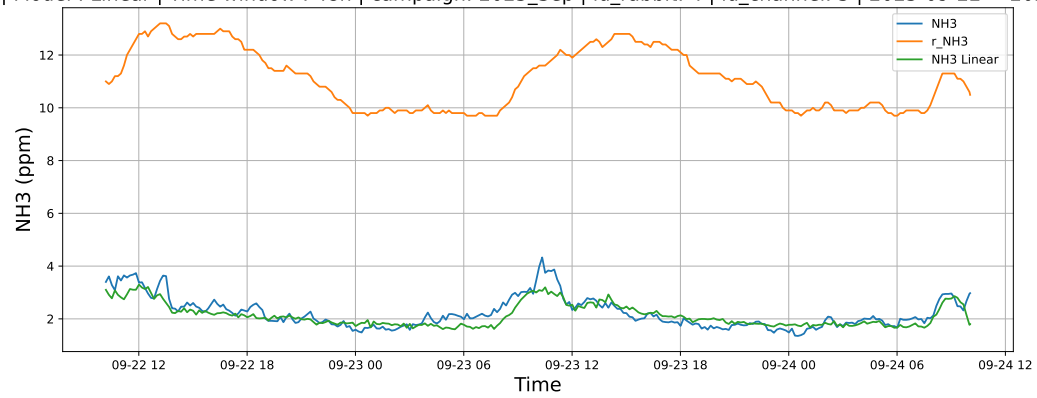


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

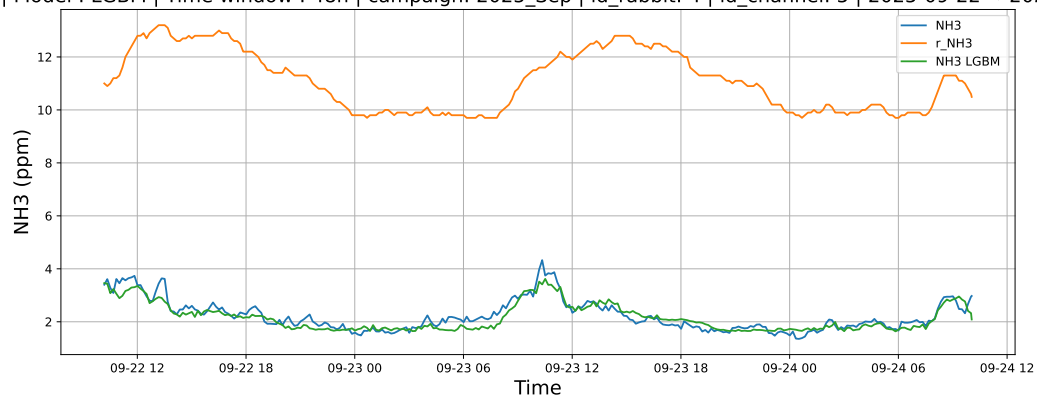


4.1.7.2 Time window : 48

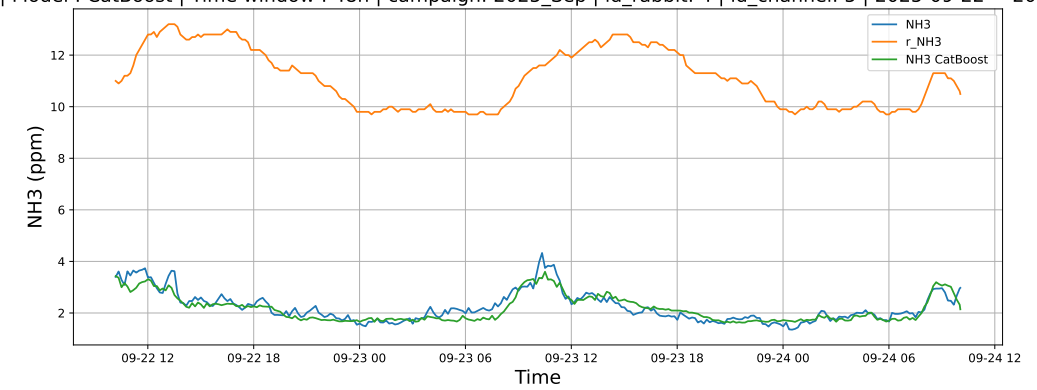
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

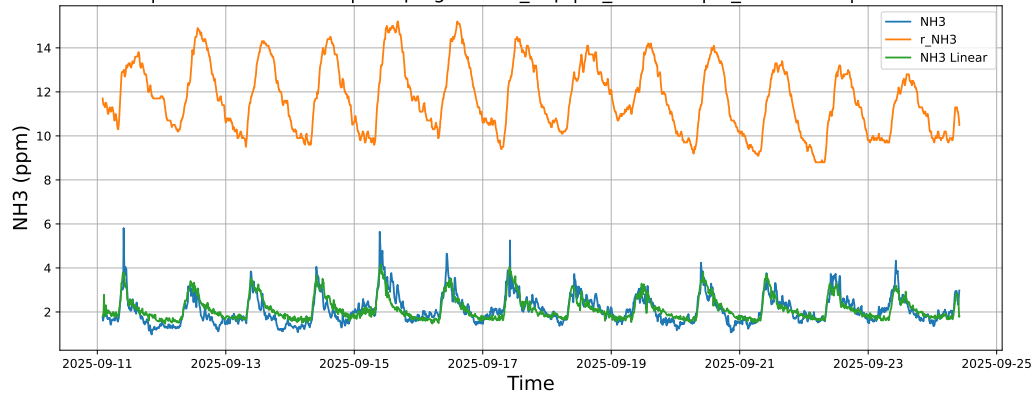


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

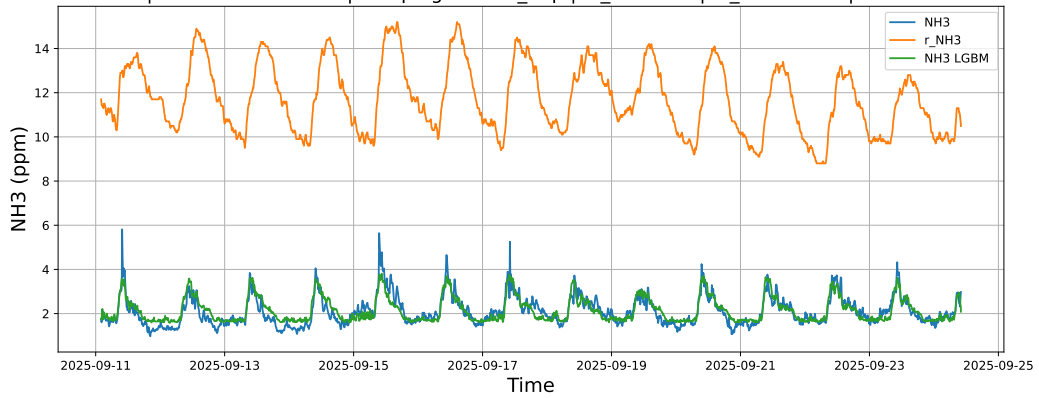


4.1.7.3 Time window : ALL

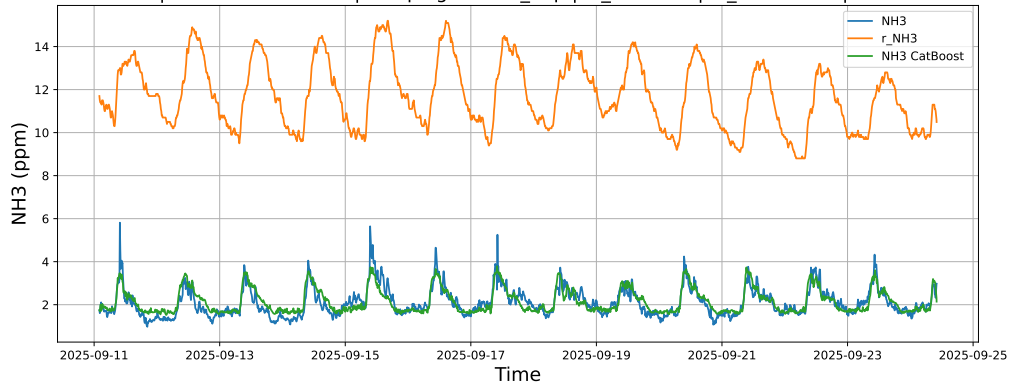
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



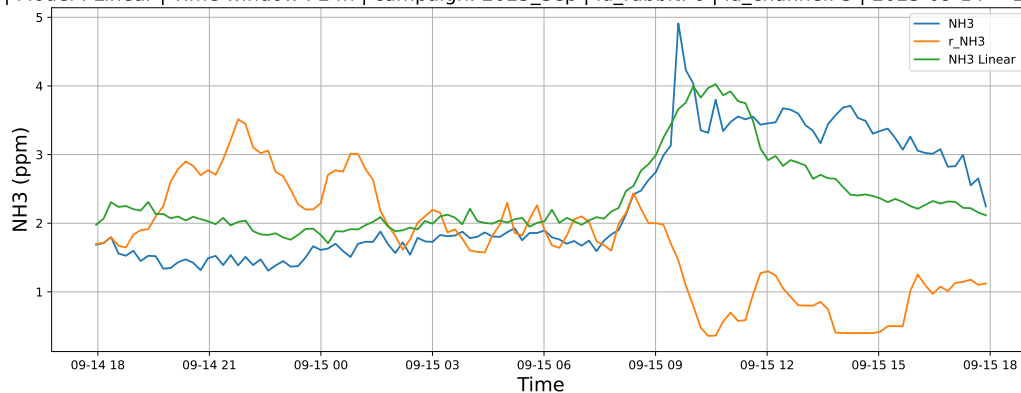
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



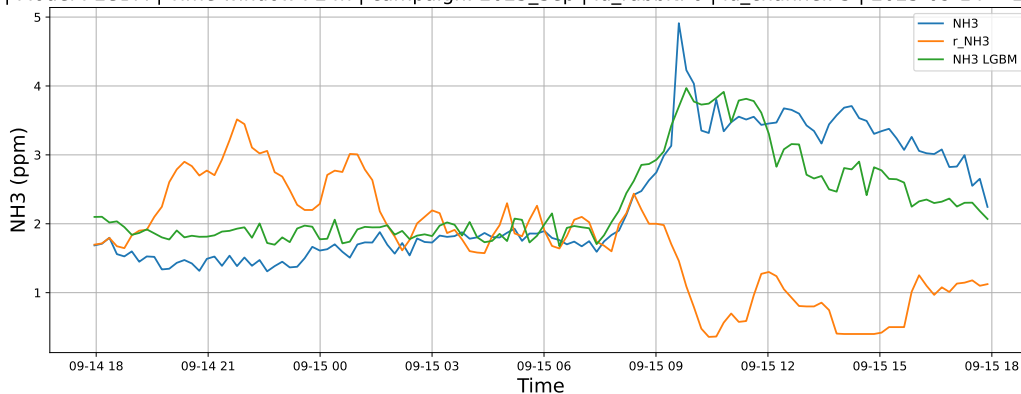
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

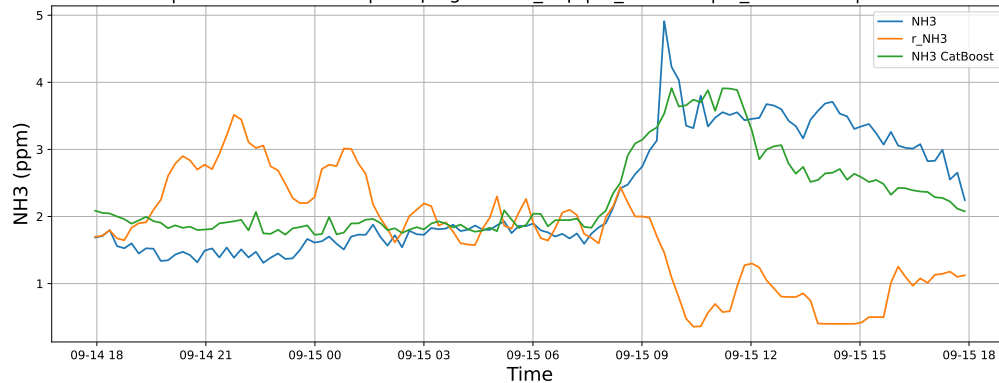
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

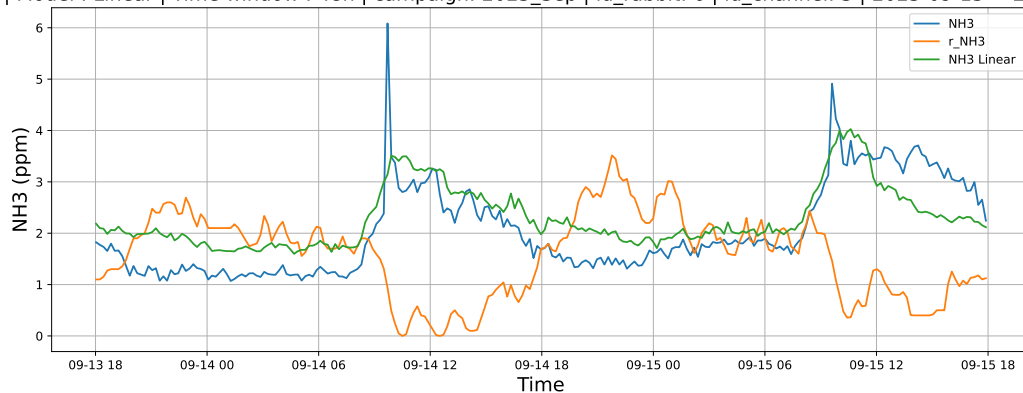


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

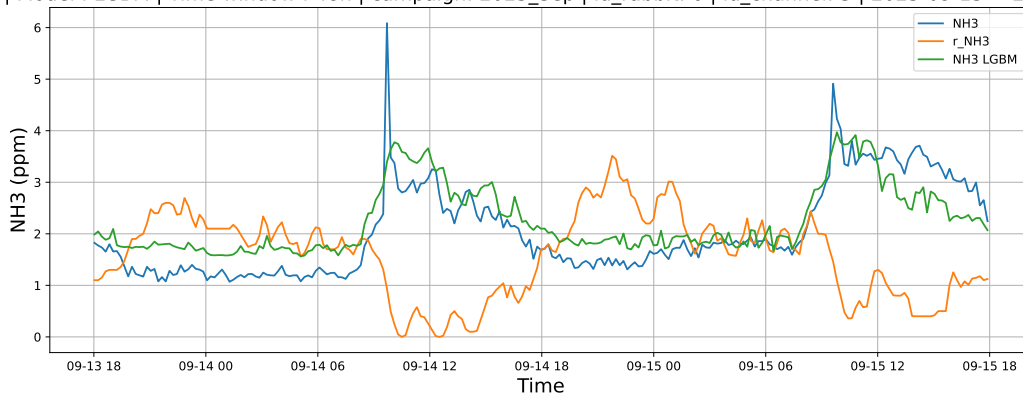


4.1.8.2 Time window : 48

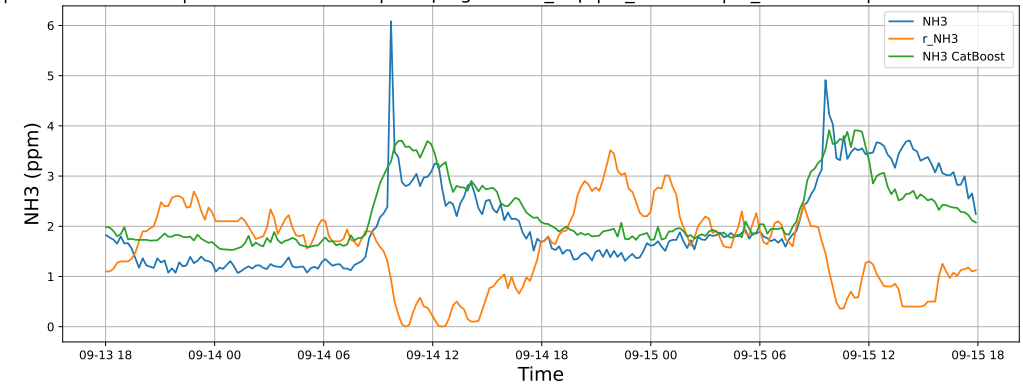
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

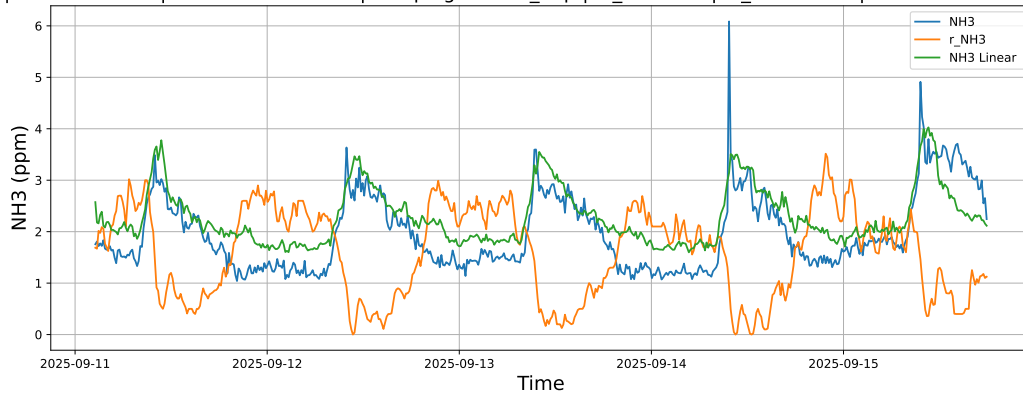


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

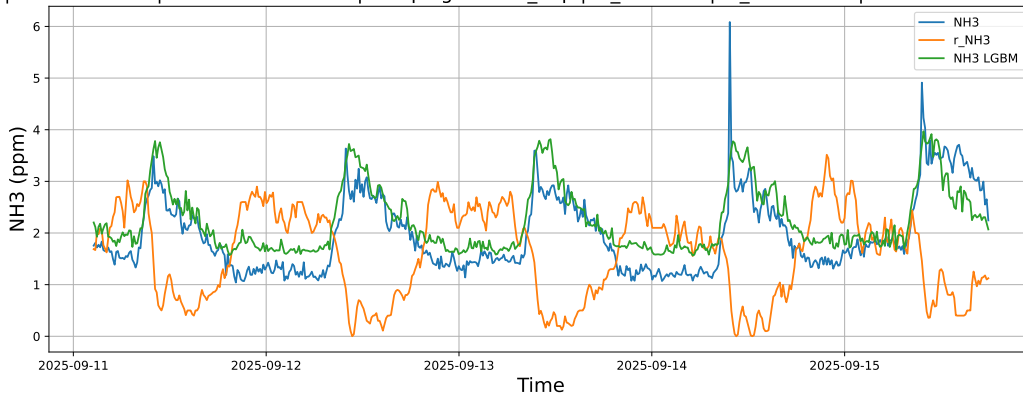


4.1.8.3 Time window : ALL

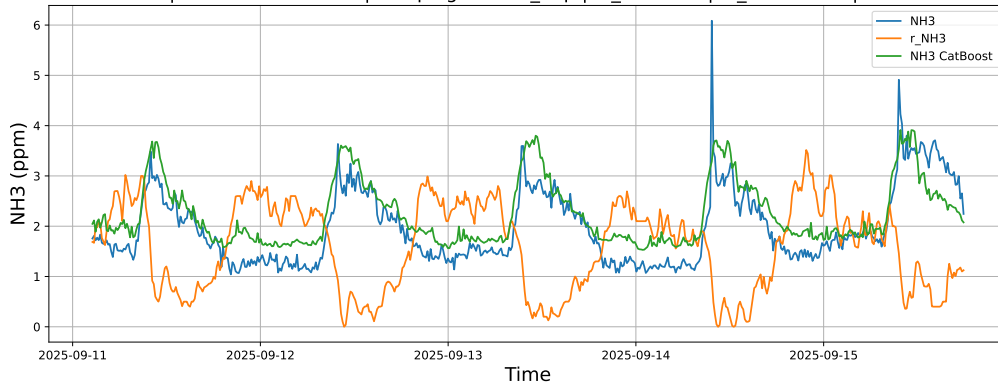
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

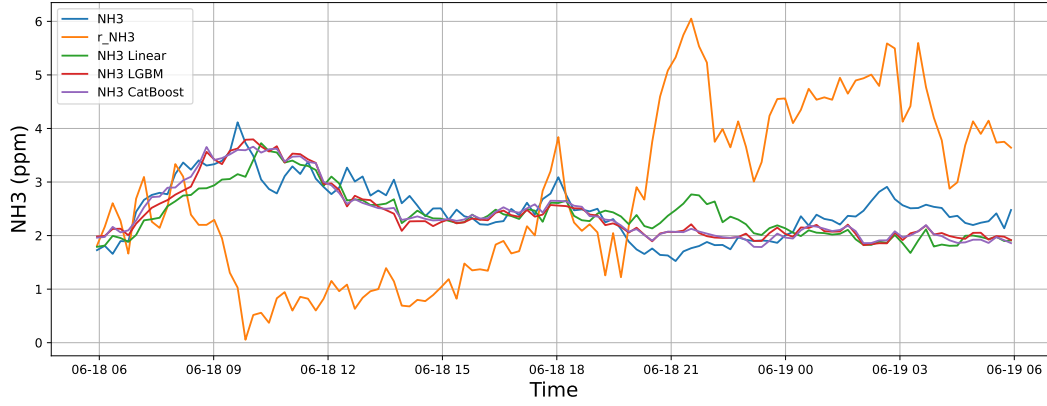


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

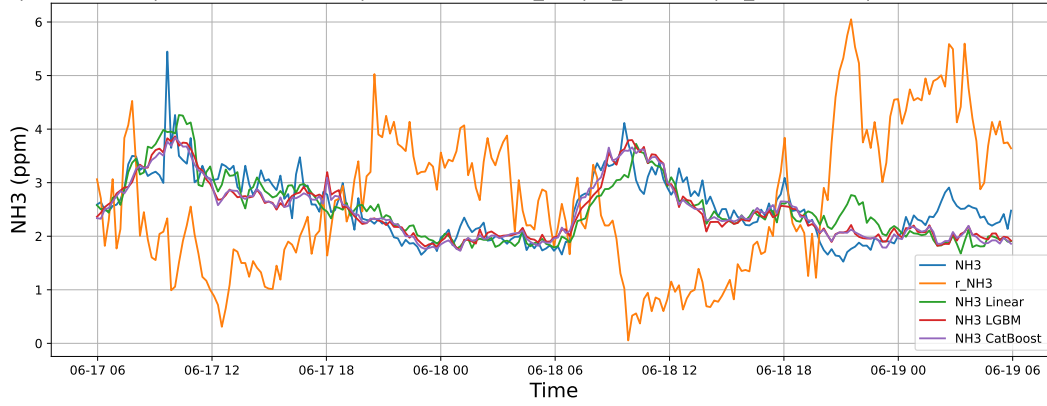
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



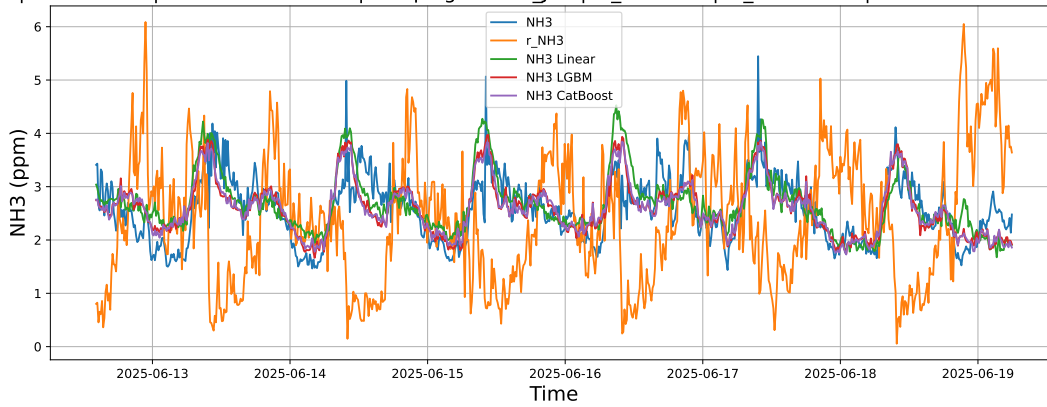
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

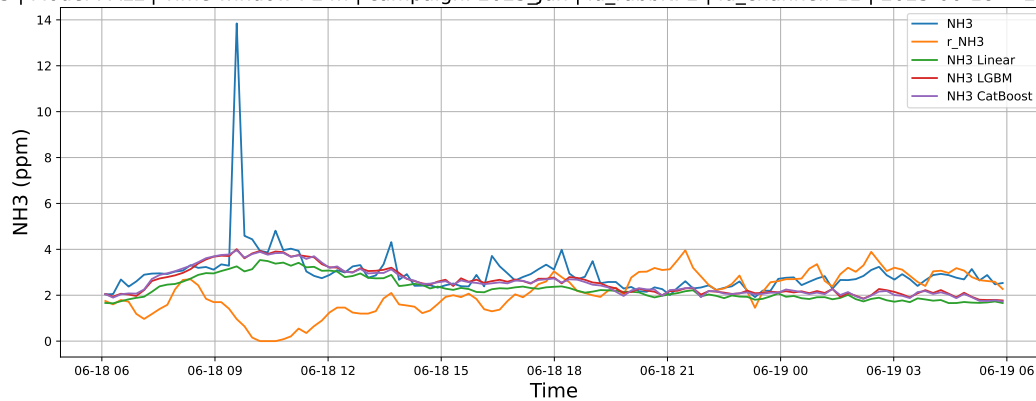
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

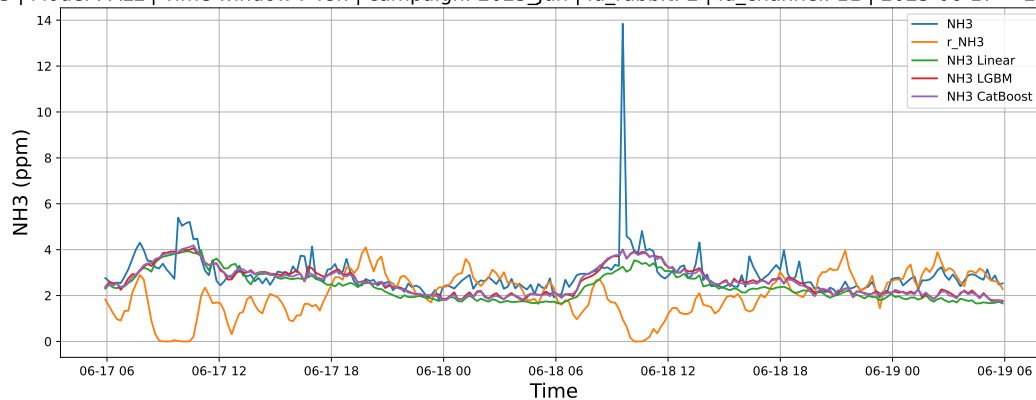
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



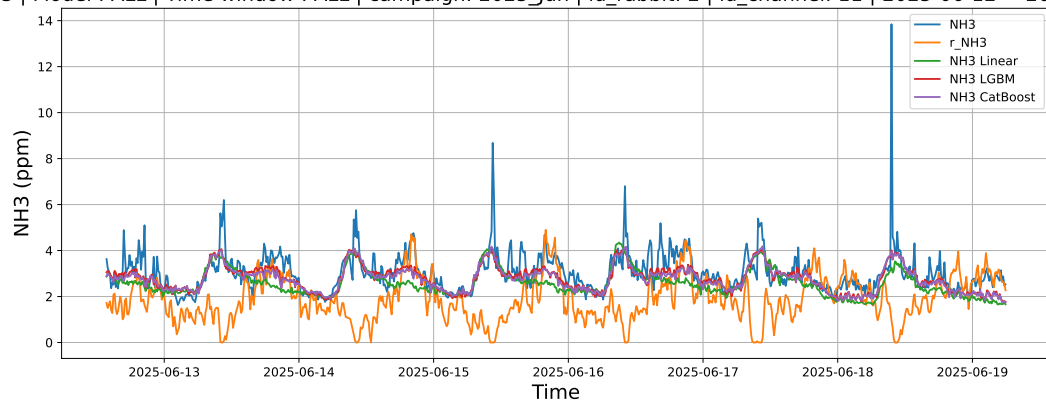
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

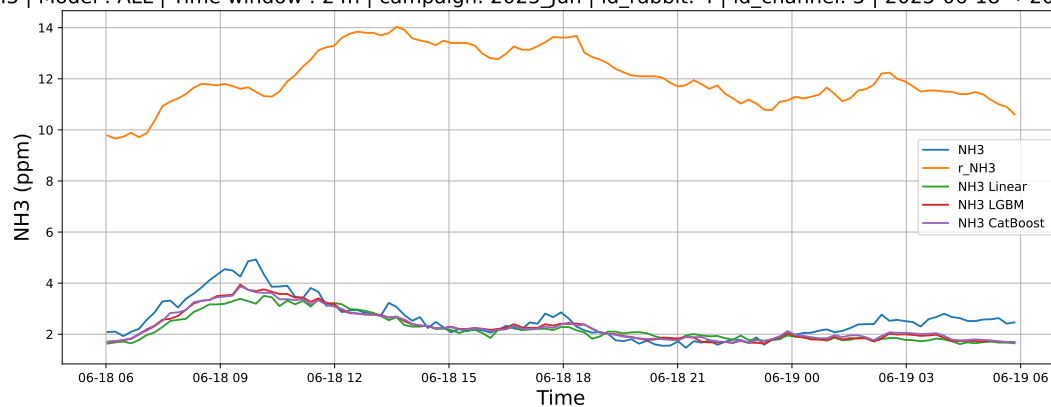
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

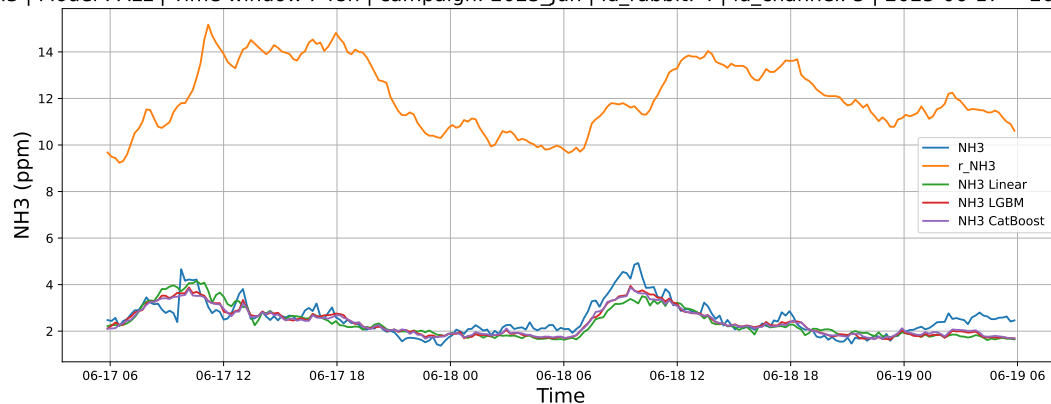
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



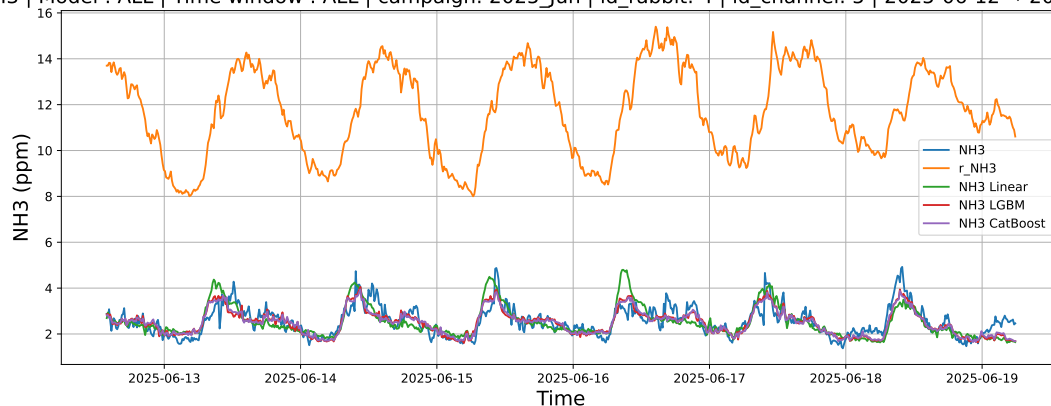
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

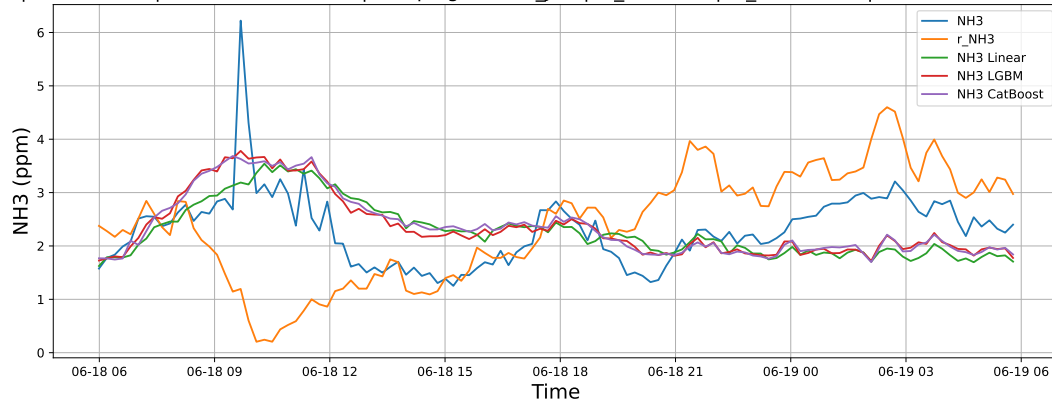
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

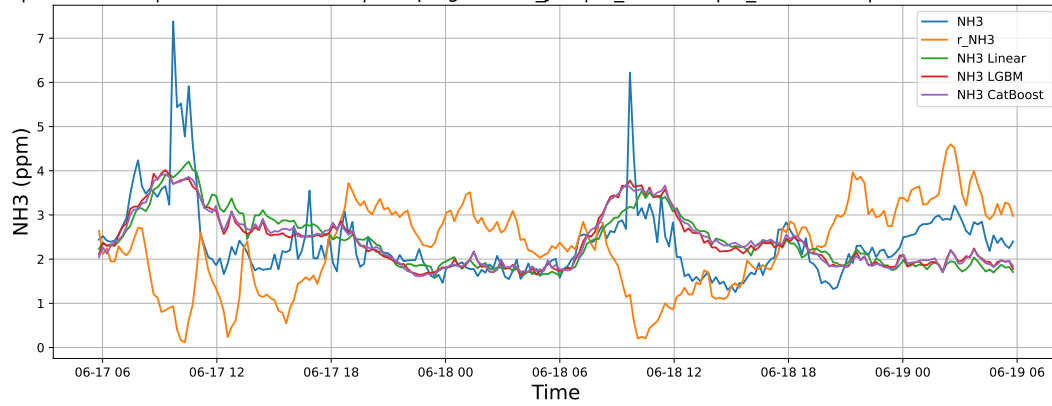
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



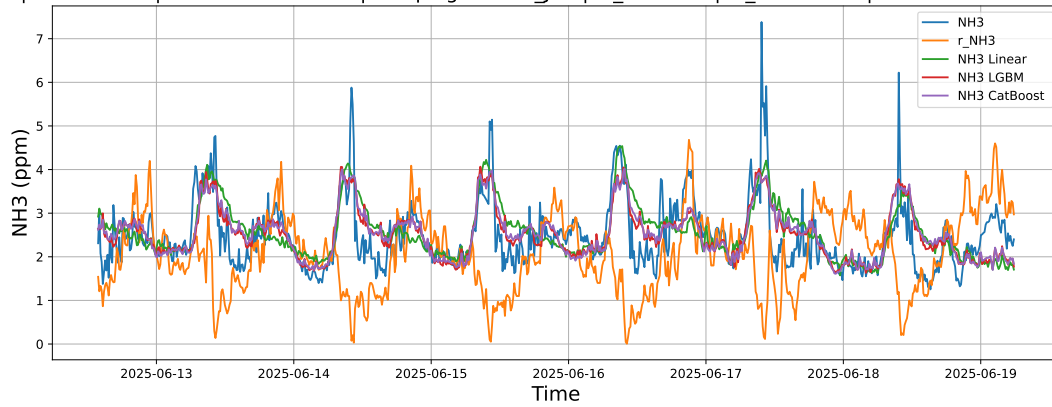
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

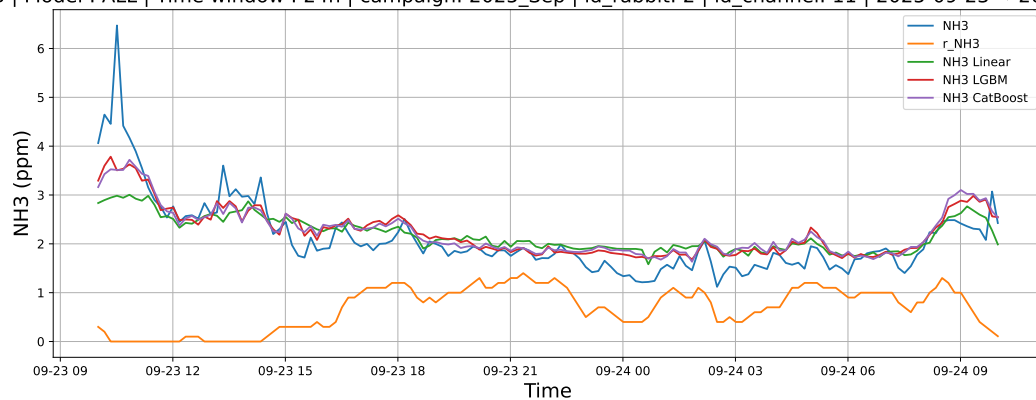
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

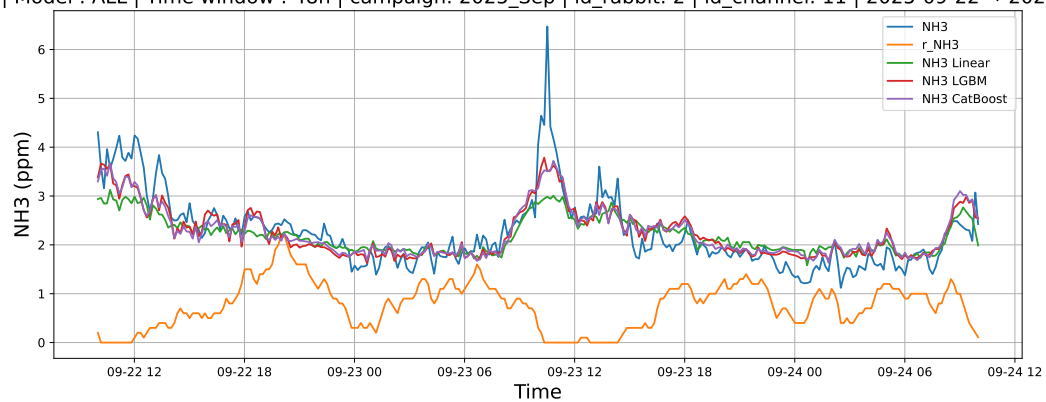
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



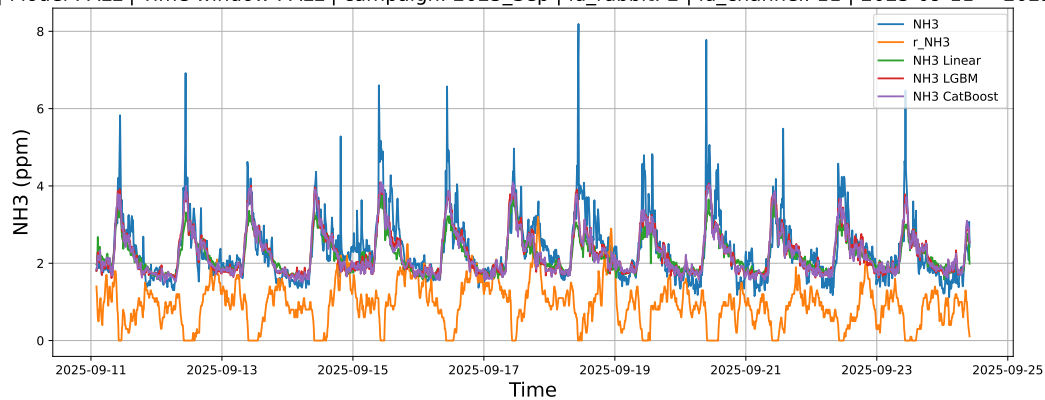
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

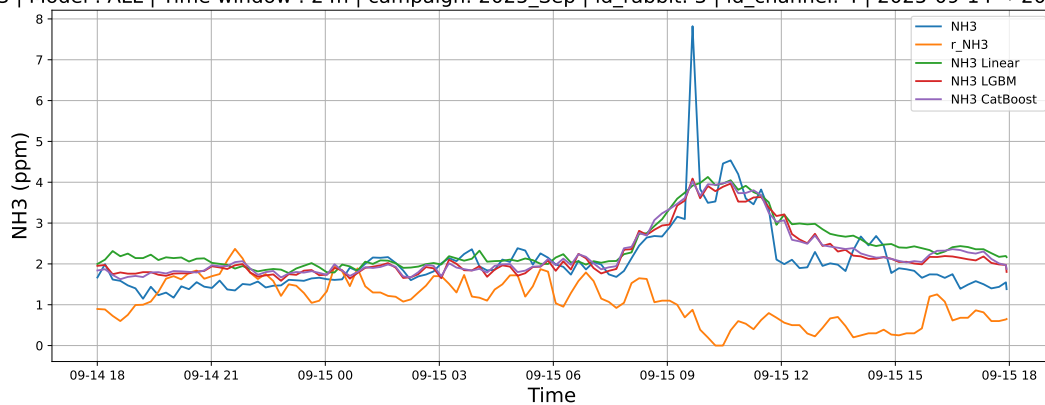
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

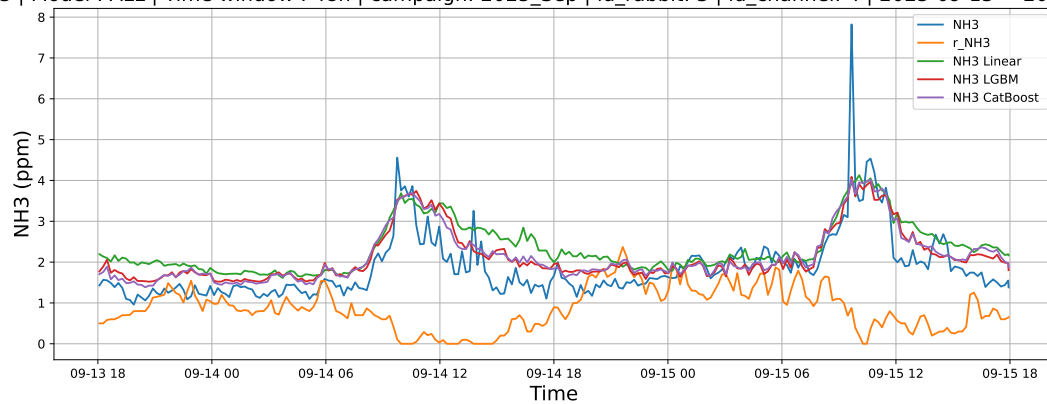
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



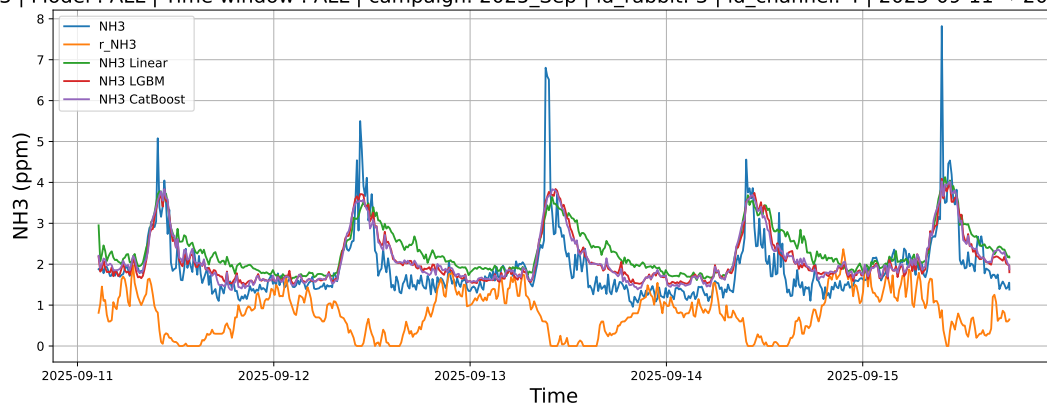
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

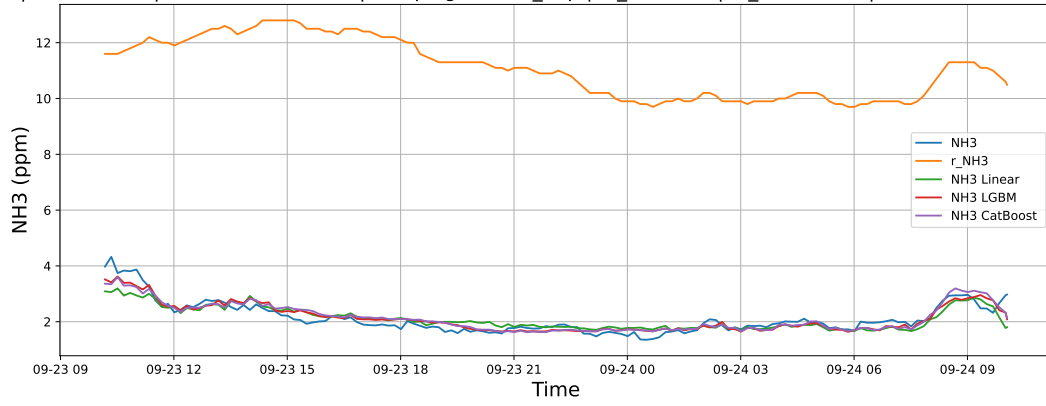
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

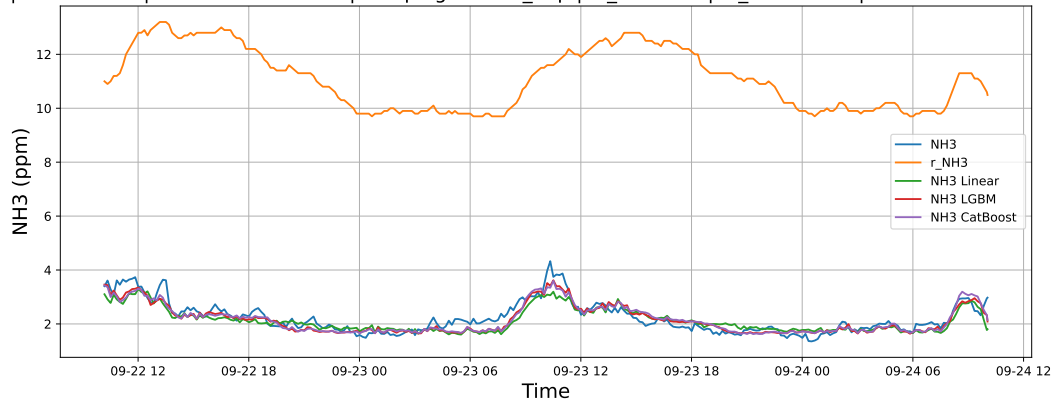
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



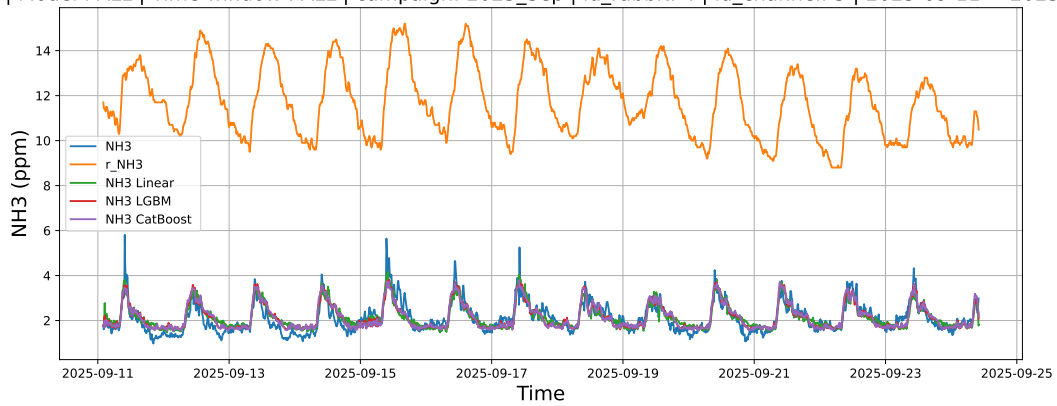
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

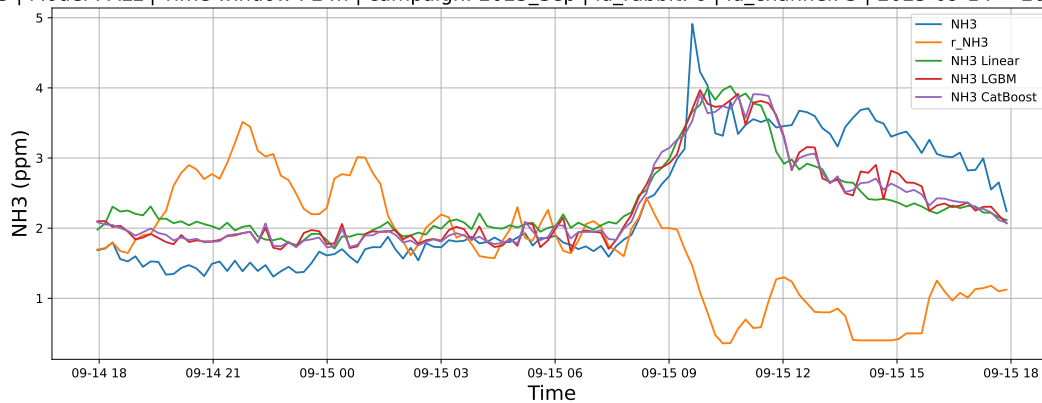
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

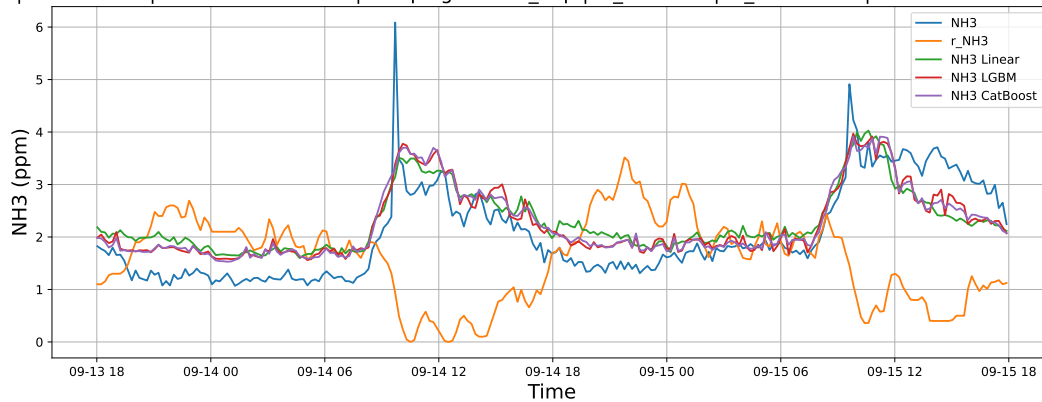
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

